Set 4C: BJT DC Bias Design

ECEN 325 — Lecture 4 (Bipolar Junction Transistors) practice set

What this set trains. Designing a bias network: choose every resistor so that the transistors carry the specified collector currents and the specified node voltages appear, while the design stays insensitive to β (I_B negligible) and to V_{BE} ($\pm 0.1\,\mathrm{V}$ changes I_C by less than 10 %). The method is the DC analysis of sets 4A/4B run backwards: fix node voltages and branch currents, then $R = \Delta V / I$ for every resistor. Each solution ends by checking the finished circuit exactly. Problems 1–6 are easy, 7–12 medium, 13–18 hard. Try each problem before looking at its solution.

The Lecture-4 pipeline (use it for every problem in this lecture)

Tools — this is all the theory you need (plus the Lecture-1–3 tools):

T1 *Active BJT at DC:* npn $V_{BE} = 0.7\,\mathrm{V}$, pnp $V_{EB} = 0.7\,\mathrm{V}$; $I_C = \beta I_B$, $I_E = (\beta+1) I_B$, so $I_C \approx I_E$ and, when β is large, $I_B \approx 0$. Active needs $|V_{CE}| \geq 0.2\,\mathrm{V}$.

T2 *Current mirror:* transistors with the same base node and the same emitter node have the same V_{BE}, hence the same I_C.

T3 *Small-signal parameters:* $r_e = \frac{V_T}{I_C}$ ($V_T = 25\,\mathrm{mV}$), $\alpha = \frac{\beta}{\beta+1}$, $r_o = \infty$. For AC, DC supplies are ground and "∞" capacitors are shorts.

T4 *Node impedances and gains:* $Z_{base} = (\beta+1)(r_e + Z_E)$, $Z_{emitter} = r_e + \frac{Z_B}{\beta+1}$, $Z_{collector} = \infty$; CE $\frac{v_c}{v_b} = -\alpha \frac{Z_C}{r_e + Z_E}$, CB $\frac{v_c}{v_e} = \alpha \frac{Z_C}{Z_{emitter}}$, CC $\frac{v_e}{v_b} = \frac{Z_E}{r_e + Z_E}$. Z_C, Z_E, Z_B are everything else hanging on that node (including the input resistance of the next stage).

Steps:

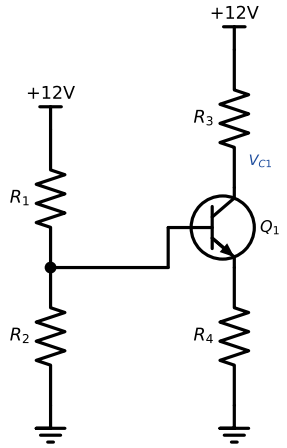
1. *DC:* assume every transistor active and $I_B \approx 0$; walk through the circuit node by node (dividers, V_{BE}, KCL for I_E, $I_C \approx I_E$, collector voltages); then *verify*: $I_B \ll$ bias currents and $|V_{CE}| \geq 0.2\,\mathrm{V}$.
2. *AC:* r_e of every transistor; input resistance of every stage from the last one backwards; multiply the node-to-node gains along the signal path; then standard form, Bode plot and output as in Lecture 1.
3. *Design:* run the same relations backwards, choosing the free quantities with the stated design rules, then check the result by analysing the finished circuit.

Problems

Easy problems

Problem 4C.1 — Easy

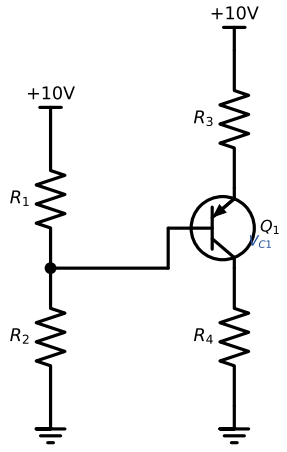
Design the circuit below (find all resistor values) for the specifications in the table, such that every I_B is negligible and $|V_{BE}| = 0.7 \pm 0.1\,\text{V}$ changes every I_C by less than 10 %. $\beta = 100$. Where the specifications leave a choice, use the smallest emitter-resistor voltage the V_{BE} rule allows, divider current $I_{R2} = 500\,\mu\text{A}$.



specification	value
I_{C1}	1 mA
V_{C1}	7 V

Problem 4C.2 — Easy

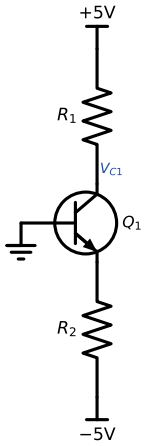
Design the circuit below (find all resistor values) for the specifications in the table, such that every I_B is negligible and $|V_{BE}| = 0.7 \pm 0.1\,\text{V}$ changes every I_C by less than 10 %. $\beta = 100$. Where the specifications leave a choice, use the smallest emitter-resistor voltage the V_{BE} rule allows, divider current $I_{R2} = 1\,\text{mA}$.



specification	value
I_{C1}	2 mA
V_{C1}	4 V

Problem 4C.3 — Easy

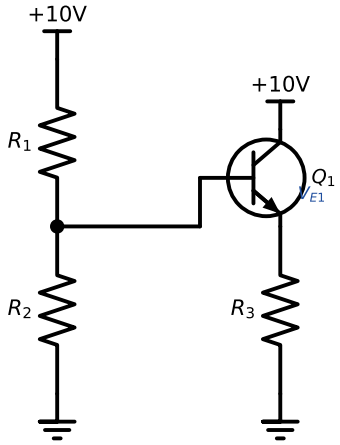
Design the circuit below (find all resistor values) for the specifications in the table, such that every I_B is negligible and $|V_{BE}| = 0.7 \pm 0.1\,\text{V}$ changes every I_C by less than 10 %. $\beta = 100$.



specification	value
I_{C1}	1 mA
V_{C1}	2 V

Problem 4C.4 — Easy

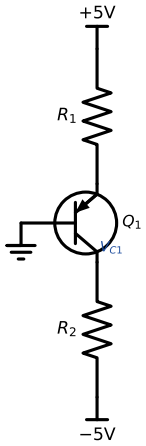
Design the circuit below (find all resistor values) for the specifications in the table, such that every I_B is negligible and $|V_{BE}| = 0.7 \pm 0.1\,\text{V}$ changes every I_C by less than 10 %. $\beta = 100$. Where the specifications leave a choice, use divider current $I_{R2} = 1\,\text{mA}$.



specification	value
I_{C1}	5 mA
V_{E1}	5 V
$\lvert V_{CE}\rvert$ (all transistors)	≥ 1 V

Problem 4C.5 — Easy

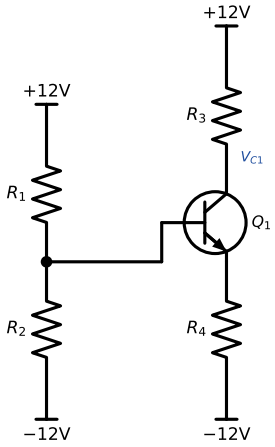
Design the circuit below (find all resistor values) for the specifications in the table, such that every I_B is negligible and $|V_{BE}| = 0.7 \pm 0.1\,\text{V}$ changes every I_C by less than 10 %. $\beta = 100$.



specification	value
I_{C1}	2 mA
V_{C1}	−1 V

Problem 4C.6 — Easy

Design the circuit below (find all resistor values) for the specifications in the table, such that every I_B is negligible and $|V_{BE}| = 0.7 \pm 0.1\,\text{V}$ changes every I_C by less than 10 %. $\beta = 100$. Where the specifications leave a choice, use the smallest emitter-resistor voltage the V_{BE} rule allows, divider current $I_{R2} = 1\,\text{mA}$.

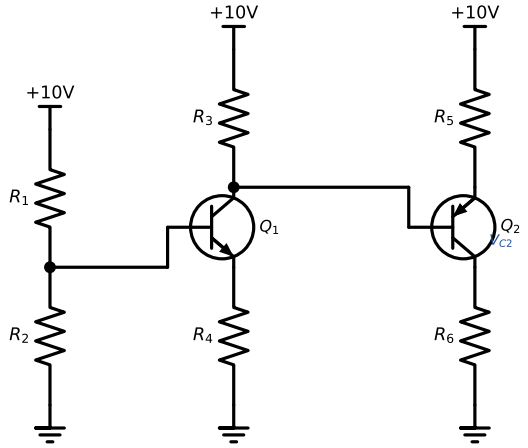


specification	value
I_{C1}	2 mA
V_{C1}	4 V

Medium problems

Problem 4C.7 — Medium

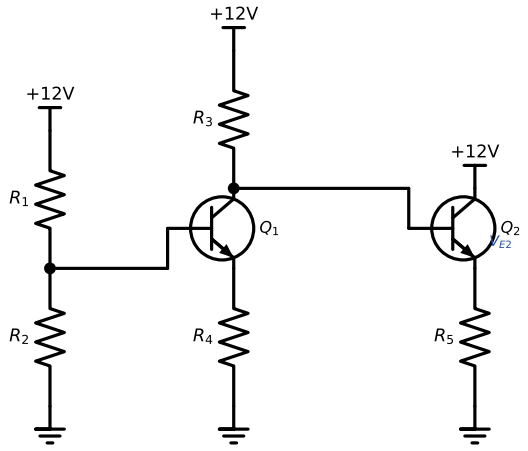
Design the circuit below (find all resistor values) for the specifications in the table, such that every I_B is negligible and $|V_{BE}| = 0.7 \pm 0.1\,\text{V}$ changes every I_C by less than 10 %. $\beta = 100$. Where the specifications leave a choice, use the smallest emitter-resistor voltage the V_{BE} rule allows, divider current $I_{R2} = 500\,\mu\text{A}$.



specification	value
I_{C1}	1 mA
I_{C2}	2 mA
V_{C2}	4 V
$\lvert V_{CE}\rvert$ (all transistors)	≥ 1 V

Problem 4C.8 — Medium

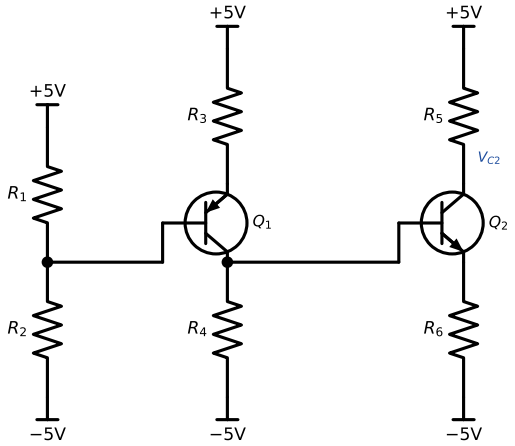
Design the circuit below (find all resistor values) for the specifications in the table, such that every I_B is negligible and $|V_{BE}| = 0.7 \pm 0.1\,\text{V}$ changes every I_C by less than 10 %. $\beta = 100$. Where the specifications leave a choice, use the smallest emitter-resistor voltage the V_{BE} rule allows, divider current $I_{R2} = 500\,\mu\text{A}$.



specification	value
I_{C1}	1 mA
I_{C2}	4 mA
V_{E2}	6 V
$\lvert V_{CE}\rvert$ (all transistors)	≥ 1 V

Problem 4C.9 — Medium

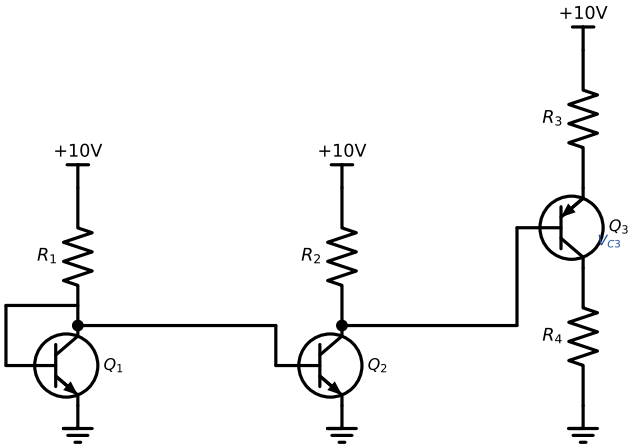
Design the circuit below (find all resistor values) for the specifications in the table, such that every I_B is negligible and $|V_{BE}| = 0.7 \pm 0.1\,\text{V}$ changes every I_C by less than 10 %. $\beta = 100$. Where the specifications leave a choice, use the smallest emitter-resistor voltage the V_{BE} rule allows, divider current $I_{R2} = 500\,\mu\text{A}$.



specification	value
I_{C1}	1 mA
I_{C2}	2 mA
V_{C2}	1 V
$\lvert V_{CE}\rvert$ (all transistors)	$\geq 1\,\text{V}$

Problem 4C.10 — Medium

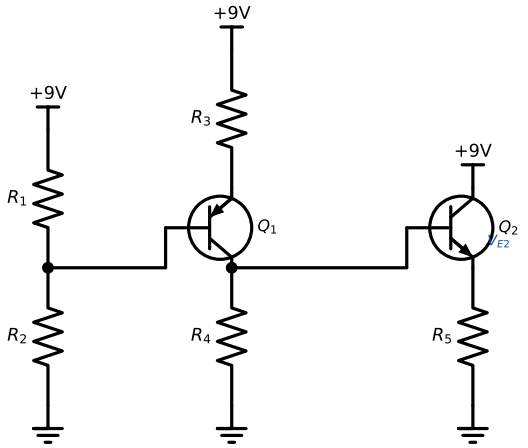
Design the circuit below (find all resistor values) for the specifications in the table, such that every I_B is negligible and $|V_{BE}| = 0.7 \pm 0.1\,\text{V}$ changes every I_C by less than 10 %. $\beta = 100$. Where the specifications leave a choice, use the smallest emitter-resistor voltage the V_{BE} rule allows.



specification	value
I_{C1}	1 mA
I_{C3}	2 mA
V_{C3}	3 V
$\lvert V_{CE}\rvert$ (all transistors)	$\geq 1\,\text{V}$

Problem 4C.11 — Medium

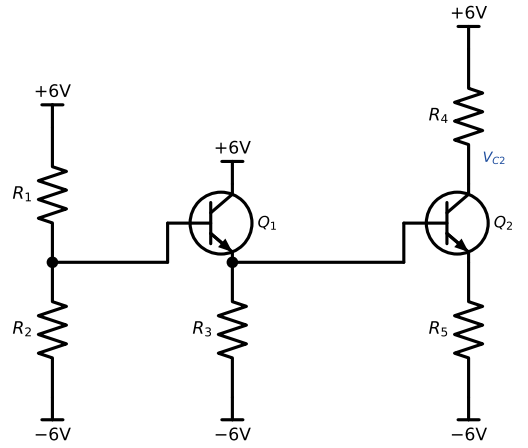
Design the circuit below (find all resistor values) for the specifications in the table, such that every I_B is negligible and $|V_{BE}| = 0.7 \pm 0.1\,\text{V}$ changes every I_C by less than 10 %. $\beta = 100$. Where the specifications leave a choice, use the smallest emitter-resistor voltage the V_{BE} rule allows, divider current $I_{R2} = 500\,\mu\text{A}$.



specification	value
I_{C1}	1 mA
I_{C2}	3 mA
V_{E2}	3 V
$\lvert V_{CE}\rvert$ (all transistors)	$\geq 1\,\text{V}$

Problem 4C.12 — Medium

Design the circuit below (find all resistor values) for the specifications in the table, such that every I_B is negligible and $|V_{BE}| = 0.7 \pm 0.1\,\text{V}$ changes every I_C by less than 10 %. $\beta = 100$. Where the specifications leave a choice, use the smallest emitter-resistor voltage the V_{BE} rule allows, divider current $I_{R2} = 500\,\mu\text{A}$.

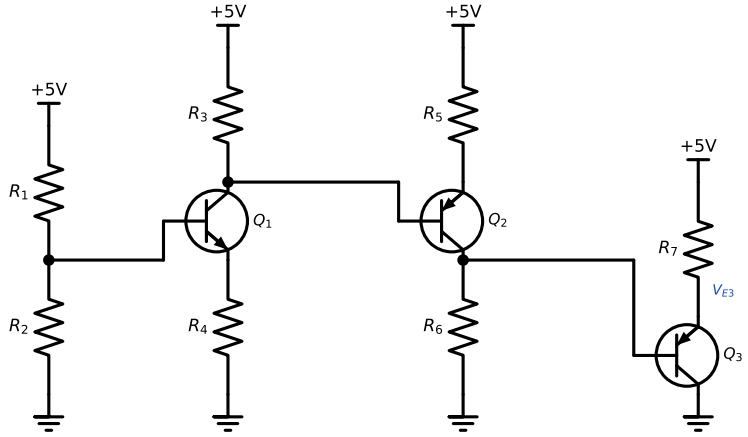


specification	value
I_{C1}	1 mA
I_{C2}	2 mA
V_{C2}	2 V
$\lvert V_{CE}\rvert$ (all transistors)	$\geq 1\,\text{V}$

Hard problems

Problem 4C.13 — Hard

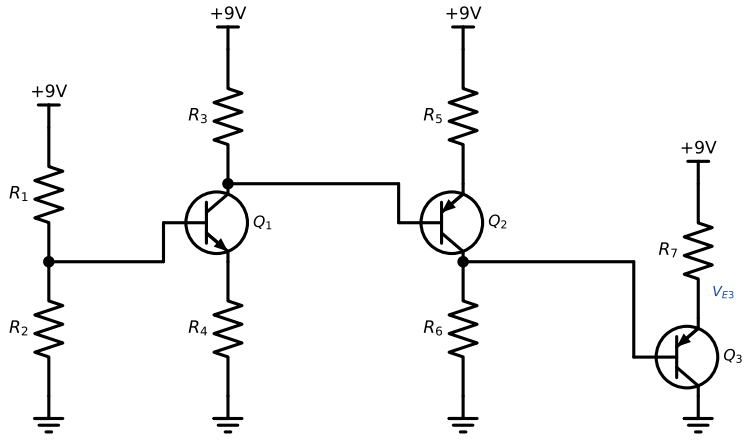
Design the circuit below (find all resistor values) for the specifications in the table, such that every I_B is negligible and $|V_{BE}| = 0.7 \pm 0.1\,\text{V}$ changes every I_C by less than 10 %. $\beta = 100$. Where the specifications leave a choice, use the smallest emitter-resistor voltage the V_{BE} rule allows. (This is Sample Problem 3 of the lecture.)



specification	value
I_{C1}	2 mA
I_{C2}	4 mA
I_{C3}	6 mA
V_{E3}	2.5 V
$\lvert V_{CE}\rvert$ (all transistors)	≥ 1 V
I_{dc} (supply current)	≤ 14 mA

Problem 4C.14 — Hard

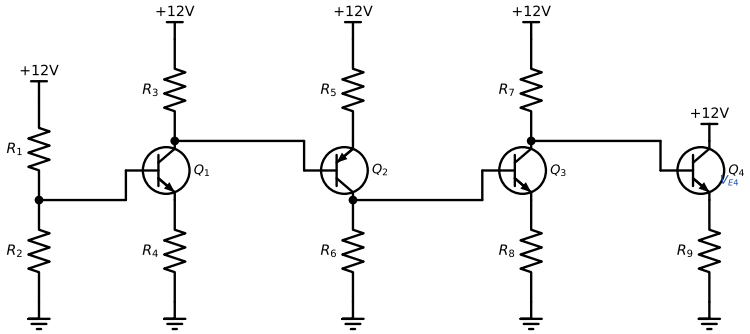
Design the circuit below (find all resistor values) for the specifications in the table, such that every I_B is negligible and $|V_{BE}| = 0.7 \pm 0.1\,\text{V}$ changes every I_C by less than 10 %. $\beta = 100$. Where the specifications leave a choice, use the smallest emitter-resistor voltage the V_{BE} rule allows.



specification	value
I_{C1}	1 mA
I_{C2}	2 mA
I_{C3}	5 mA
V_{E3}	5 V
$\lvert V_{CE}\rvert$ (all transistors)	≥ 1 V
I_{dc} (supply current)	≤ 9 mA

Problem 4C.15 — Hard

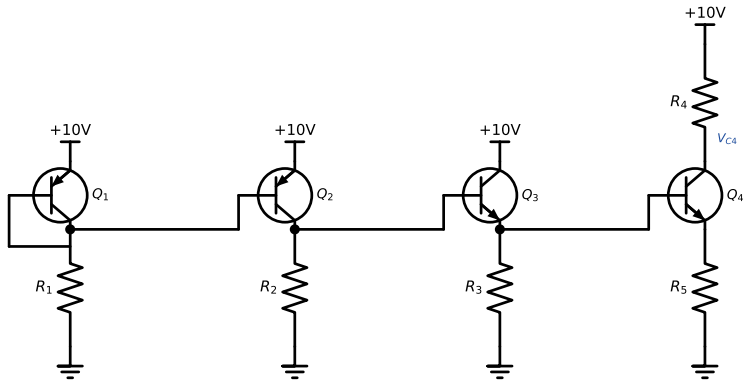
Design the circuit below (find all resistor values) for the specifications in the table, such that every I_B is negligible and $|V_{BE}| = 0.7 \pm 0.1\,\text{V}$ changes every I_C by less than 10 %. $\beta = 100$. Where the specifications leave a choice, use the smallest emitter-resistor voltage the V_{BE} rule allows, $V_{R5} = 2\,\text{V}$, $V_{R8} = 2\,\text{V}$, divider current $I_{R2} = 500\,\mu\text{A}$.



specification	value
I_{C1}	1 mA
I_{C2}	2 mA
I_{C3}	2 mA
I_{C4}	5 mA
V_{E4}	6 V
$\lvert V_{CE}\rvert$ (all transistors)	$\geq 1\,\text{V}$

Problem 4C.16 — Hard

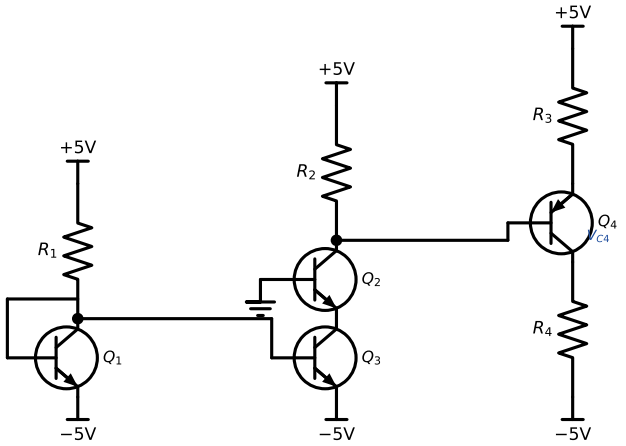
Design the circuit below (find all resistor values) for the specifications in the table, such that every I_B is negligible and $|V_{BE}| = 0.7 \pm 0.1\,\text{V}$ changes every I_C by less than 10 %. $\beta = 100$.



specification	value
I_{C1}	2 mA
I_{C3}	4 mA
I_{C4}	1 mA
V_{E3}	4 V
V_{C4}	7 V
$\lvert V_{CE}\rvert$ (all transistors)	$\geq 1\,\text{V}$
I_{dc} (supply current)	$\leq 10\,\text{mA}$

Problem 4C.17 — Hard

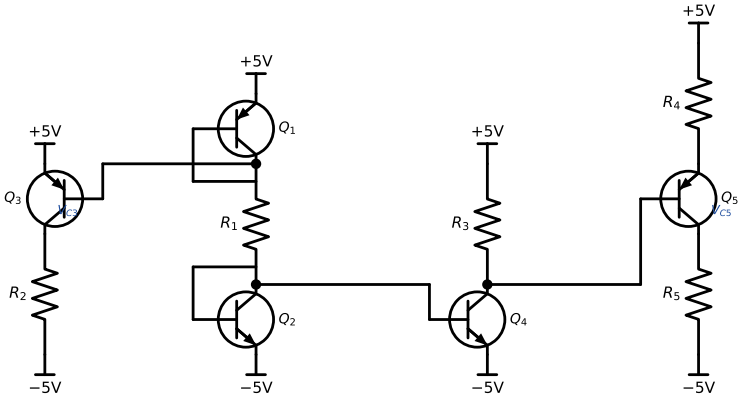
Design the circuit below (find all resistor values) for the specifications in the table, such that every I_B is negligible and $|V_{BE}| = 0.7 \pm 0.1\,\text{V}$ changes every I_C by less than 10 %. $\beta = 100$. Where the specifications leave a choice, use the smallest emitter-resistor voltage the V_{BE} rule allows.



specification	value
I_{C1}	1 mA
I_{C4}	2 mA
V_{C4}	−1 V
$\lvert V_{CE}\rvert$ (all transistors)	$\geq$ 1 V

Problem 4C.18 — Hard

Design the circuit below (find all resistor values) for the specifications in the table, such that every I_B is negligible and $|V_{BE}| = 0.7 \pm 0.1\,\text{V}$ changes every I_C by less than 10 %. $\beta = 100$. Where the specifications leave a choice, use the smallest emitter-resistor voltage the V_{BE} rule allows.

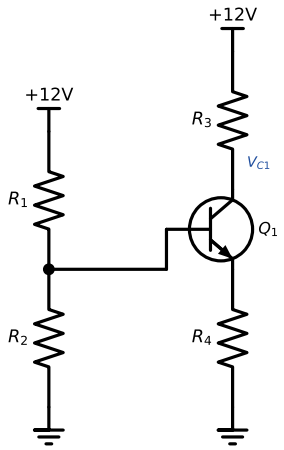


specification	value
I_{C1}	1 mA
I_{C5}	2 mA
V_{C3}	2 V
V_{C5}	−2 V
$\lvert V_{CE}\rvert$ (all transistors)	$\geq$ 1 V

Solutions

Every solution follows the steps of the Lecture-4 pipeline card.

Solution 4C.1 — Easy



Step 1 — Rules

With $I_B \approx 0$ and every transistor active ($|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$), a resistor is known as soon as the voltages at both of its ends and the current through it are known: $R = \Delta V / I$. The specifications give the currents and some node voltages; the remaining freedom is removed with two rules:

- V_{BE}-insensitive: the voltage across the resistor that sets an emitter current must be at least 1 V (then $\pm 0.1\,\text{V}$ of V_{BE} changes I_C by at most 10 %);
- β-insensitive: a divider current must be much larger ($\geq 10\times$) than the base current it feeds, and a collector current that drives a base must be much larger than that base current.

Work from the nodes that are already fixed towards the unknown ones.

Step 2 — Design, node by node

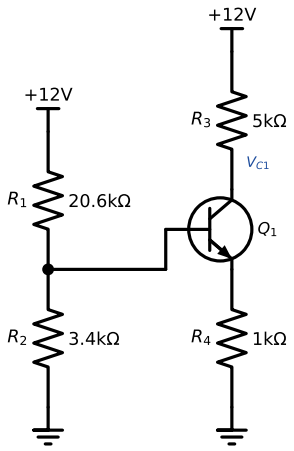
1. V_{BE}-insensitivity: a 0.1 V change of V_{BE} changes the voltage across R_4 by 0.1 V, so $\frac{\Delta I_C}{I_C} = \frac{0.1}{V_{R4}} \leq 10\%$ needs $V_{R4} \geq 1\,\text{V}$. Choose the smallest value, $V_{R4} = 1\,\text{V}$: $V_{E1} = 0 + V_{R4} = 0 + 1 = 1\,\text{V}$.
2. The divider current must be much larger than the base current it feeds ($I_{B1} = 10\,\mu\text{A}$). Choose $I_{R2} = 500\,\mu\text{A}$ (50 times larger).
3. Q_1 is active: $V_{B1} = V_{E1} + 0.7 = 1.7\,\text{V}$.
4. With $I_B \approx 0$ the only current through R_1 is I_{R2}: $I_{R1} = I_{R2} = 500\,\mu\text{A}$.
5. $R_1 = \dfrac{12 - V_{B1}}{I_{R2}} = \dfrac{12 - 1.7}{500\,\mu\text{A}} = \mathbf{20.6\,k\Omega}$
6. $R_2 = \dfrac{V_{B1} - 0}{I_{R2}} = \dfrac{1.7 - 0}{500\,\mu\text{A}} = \mathbf{3.4\,k\Omega}$
7. With $I_B \approx 0$ the only current through R_3 is I_{C1}: $I_{R3} = I_{C1} = 1\,\text{mA}$.
8. $R_3 = \dfrac{12 - V_{C1}}{I_{C1}} = \dfrac{12 - 7}{1\,\text{mA}} = \mathbf{5\,k\Omega}$

9. With $I_B \approx 0$ the only current through R_4 is I_{E1}: $I_{R4} = I_{E1} = 1\,\text{mA}$ ($I_C \approx I_E$).

10. $R_4 = \dfrac{V_{E1} - 0}{I_{E1}} = \dfrac{1 - 0}{1\,\text{mA}} = \mathbf{1\,k\Omega}$

Step 3 — Designed circuit

resistor	value
R_1	$20.6\,\text{k}\Omega$
R_2	$3.4\,\text{k}\Omega$
R_3	$5\,\text{k}\Omega$
R_4	$1\,\text{k}\Omega$



Step 4 — Check the design

- $I_{B1} = \frac{I_{C1}}{\beta} = 10\,\mu\text{A} \ll$ the $500\,\mu\text{A}$ flowing in the base network ✓
- $V_{CE1} = V_{C1} - V_{E1} = 7 - 1 = 6\,\text{V} \geq 0.2\,\text{V}$ ✓

Finally the designed circuit is solved exactly ($\beta = 100$, base currents included), once with $|V_{BE}| = 0.7\,\text{V}$ and once with each transistor's $|V_{BE}|$ changed to 0.6 and 0.8 V:

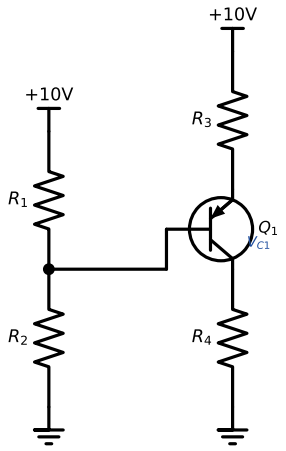
	target I_C	exact I_C	$\lvert V_{BE}\rvert = 0.6$	$\lvert V_{BE}\rvert = 0.8$	max change
Q_1	1 mA	$962\,\mu\text{A}$	1.06 mA	$866\,\mu\text{A}$	9.6 %

A $\pm 0.1\,\text{V}$ change of V_{BE} moves no current by more than about 10 % of its design value (V_{BE}-insensitive), and the exact currents are close to the targets (I_B negligible).

Answer

$R_1 = 20.6\,\text{k}\Omega, \quad R_2 = 3.4\,\text{k}\Omega, \quad R_3 = 5\,\text{k}\Omega, \quad R_4 = 1\,\text{k}\Omega$

Solution 4C.2 **Easy**



Step 1 — Rules

With $I_B \approx 0$ and every transistor active ($|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$), a resistor is known as soon as the voltages at both of its ends and the current through it are known: $R = \Delta V / I$. The specifications give the currents and some node voltages; the remaining freedom is removed with two rules:

- V_{BE}-insensitive: the voltage across the resistor that sets an emitter current must be at least 1 V (then $\pm 0.1\,\text{V}$ of V_{BE} changes I_C by at most 10 %);
- β-insensitive: a divider current must be much larger ($\geq 10\times$) than the base current it feeds, and a collector current that drives a base must be much larger than that base current.

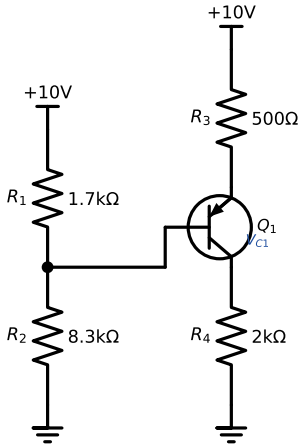
Work from the nodes that are already fixed towards the unknown ones.

Step 2 — Design, node by node

1. V_{BE}-insensitivity: a 0.1 V change of V_{BE} changes the voltage across R_3 by 0.1 V, so $\frac{\Delta I_C}{I_C} = \frac{0.1}{V_{R3}} \leq 10\%$ needs $V_{R3} \geq 1\,\text{V}$. Choose the smallest value, $V_{R3} = 1\,\text{V}$: $V_{E1} = 10 - V_{R3} = 10 - 1 = 9\,\text{V}$.
2. The divider current must be much larger than the base current it feeds ($I_{B1} = 20\,\mu\text{A}$). Choose $I_{R2} = 1\,\text{mA}$ (50 times larger).
3. Q_1 is active: $V_{B1} = V_{E1} - 0.7 = 8.3\,\text{V}$.
4. With $I_B \approx 0$ the only current through R_1 is I_{R2}: $I_{R1} = I_{R2} = 1\,\text{mA}$.
5. $R_1 = \dfrac{10 - V_{B1}}{I_{R2}} = \dfrac{10 - 8.3}{1\,\text{mA}} = \mathbf{1.7\,k\Omega}$
6. $R_2 = \dfrac{V_{B1} - 0}{I_{R2}} = \dfrac{8.3 - 0}{1\,\text{mA}} = \mathbf{8.3\,k\Omega}$
7. With $I_B \approx 0$ the only current through R_3 is I_{E1}: $I_{R3} = I_{E1} = 2\,\text{mA}$ ($I_C \approx I_E$).
8. $R_3 = \dfrac{10 - V_{E1}}{I_{E1}} = \dfrac{10 - 9}{2\,\text{mA}} = \mathbf{500\,\Omega}$
9. With $I_B \approx 0$ the only current through R_4 is I_{C1}: $I_{R4} = I_{C1} = 2\,\text{mA}$.
10. $R_4 = \dfrac{V_{C1} - 0}{I_{C1}} = \dfrac{4 - 0}{2\,\text{mA}} = \mathbf{2\,k\Omega}$

Step 3 — Designed circuit

resistor	value
R_1	$1.7\,\mathrm{k\Omega}$
R_2	$8.3\,\mathrm{k\Omega}$
R_3	$500\,\Omega$
R_4	$2\,\mathrm{k\Omega}$



Step 4 — Check the design

- $I_{B1} = \frac{I_{C1}}{\beta} = 20\,\mu\mathrm{A} \ll$ the 1 mA flowing in the base network ✓
- $V_{EC1} = V_{E1} - V_{C1} = 9 - 4 = 5\,\mathrm{V} \geq 0.2\,\mathrm{V}$ ✓

Finally the designed circuit is solved exactly ($\beta = 100$, base currents included), once with $|V_{BE}| = 0.7\,\mathrm{V}$ and once with each transistor's $|V_{BE}|$ changed to 0.6 and 0.8 V:

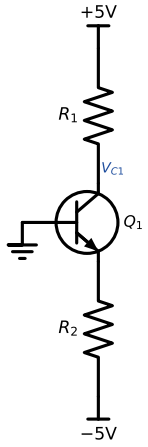
	target I_C	exact I_C	$\lvert V_{BE}\rvert = 0.6$	$\lvert V_{BE}\rvert = 0.8$	max change
Q_1	2 mA	1.93 mA	2.12 mA	1.73 mA	9.6 %

A $\pm 0.1\,\mathrm{V}$ change of V_{BE} moves no current by more than about 10 % of its design value (V_{BE}-insensitive), and the exact currents are close to the targets (I_B negligible).

Answer

$R_1 = 1.7\,\mathrm{k\Omega}, \quad R_2 = 8.3\,\mathrm{k\Omega}, \quad R_3 = 500\,\Omega, \quad R_4 = 2\,\mathrm{k\Omega}$

Solution 4C.3 **Easy**



Step 1 — Rules

With $I_B \approx 0$ and every transistor active ($|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$), a resistor is known as soon as the voltages at both of its ends and the current through it are known: $R = \Delta V / I$. The specifications give the currents and some node voltages; the remaining freedom is removed with two rules:

- V_{BE}-insensitive: the voltage across the resistor that sets an emitter current must be at least 1 V (then $\pm 0.1\,\text{V}$ of V_{BE} changes I_C by at most 10 %);
- β-insensitive: a divider current must be much larger ($\geq 10\times$) than the base current it feeds, and a collector current that drives a base must be much larger than that base current.

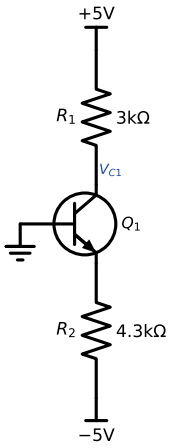
Work from the nodes that are already fixed towards the unknown ones.

Step 2 — Design, node by node

1. Q_1 is active: $V_{E1} = 0 - 0.7 = -0.7\,\text{V}$.
2. With $I_B \approx 0$ the only current through R_1 is I_{C1}: $I_{R1} = I_{C1} = 1\,\text{mA}$.
3. $R_1 = \dfrac{5 - V_{C1}}{I_{C1}} = \dfrac{5-2}{1\,\text{mA}} = \mathbf{3\,k\Omega}$
4. With $I_B \approx 0$ the only current through R_2 is I_{E1}: $I_{R2} = I_{E1} = 1\,\text{mA}$ ($I_C \approx I_E$).
5. $R_2 = \dfrac{V_{E1} - (-5)}{I_{E1}} = \dfrac{-0.7 - (-5)}{1\,\text{mA}} = \mathbf{4.3\,k\Omega}$

Step 3 — Designed circuit

resistor	value
R_1	$3\,\text{k}\Omega$
R_2	$4.3\,\text{k}\Omega$



Step 4 — Check the design

- $V_{CE1} = V_{C1} - V_{E1} = 2 - (-0.7) = 2.7\,\text{V} \geq 0.2\,\text{V}$ ✓

Finally the designed circuit is solved exactly ($\beta = 100$, base currents included), once with $|V_{BE}| = 0.7\,\text{V}$ and once with each transistor's $|V_{BE}|$ changed to 0.6 and 0.8 V:

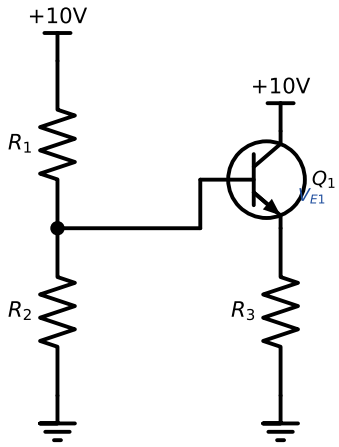
	target I_C	exact I_C	$\lvert V_{BE}\rvert = 0.6$	$\lvert V_{BE}\rvert = 0.8$	max change
Q_1	1 mA	990 μA	1.01 mA	967 μA	2.3 %

A $\pm 0.1\,\text{V}$ change of V_{BE} moves no current by more than about 10 % of its design value (V_{BE}-insensitive), and the exact currents are close to the targets (I_B negligible).

Answer

$R_1 = 3\,\text{k}\Omega, \quad R_2 = 4.3\,\text{k}\Omega$

Solution 4C.4 **Easy**



Step 1 — Rules

With $I_B \approx 0$ and every transistor active ($|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$), a resistor is known as soon as the voltages at both of its ends and the current through it are known: $R = \Delta V / I$. The specifications give the currents and some node voltages; the remaining freedom is removed with two rules:

- V_{BE}-insensitive: the voltage across the resistor that sets an emitter current must be at least 1 V (then $\pm 0.1\,\text{V}$ of V_{BE} changes I_C by at most 10 %);
- β-insensitive: a divider current must be much larger ($\geq 10\times$) than the base current it feeds, and a collector current that drives a base must be much larger than that base current.

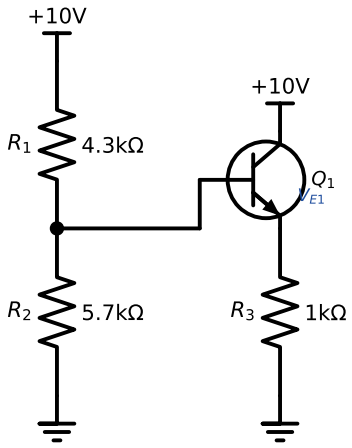
Work from the nodes that are already fixed towards the unknown ones.

Step 2 — Design, node by node

1. The divider current must be much larger than the base current it feeds ($I_{B1} = 50\,\mu\text{A}$). Choose $I_{R2} = 1\,\text{mA}$ (20 times larger).
2. Q_1 is active: $V_{B1} = V_{E1} + 0.7 = 5.7\,\text{V}$.
3. With $I_B \approx 0$ the only current through R_1 is I_{R2}: $I_{R1} = I_{R2} = 1\,\text{mA}$.
4. $R_1 = \dfrac{10 - V_{B1}}{I_{R2}} = \dfrac{10 - 5.7}{1\,\text{mA}} = \mathbf{4.3\,k\Omega}$
5. $R_2 = \dfrac{V_{B1} - 0}{I_{R2}} = \dfrac{5.7 - 0}{1\,\text{mA}} = \mathbf{5.7\,k\Omega}$
6. With $I_B \approx 0$ the only current through R_3 is I_{E1}: $I_{R3} = I_{E1} = 5\,\text{mA}$ ($I_C \approx I_E$).
7. $R_3 = \dfrac{V_{E1} - 0}{I_{E1}} = \dfrac{5 - 0}{5\,\text{mA}} = \mathbf{1\,k\Omega}$

Step 3 — Designed circuit

resistor	value
R_1	$4.3\,\mathrm{k\Omega}$
R_2	$5.7\,\mathrm{k\Omega}$
R_3	$1\,\mathrm{k\Omega}$



Step 4 — Check the design

- $I_{B1} = \frac{I_{C1}}{\beta} = 50\,\mu\mathrm{A} \ll$ the 1 mA flowing in the base network ✓
- $V_{CE1} = 10 - V_{E1} = 10 - 5 = 5\,\mathrm{V} \geq 1\,\mathrm{V}$ ✓

Finally the designed circuit is solved exactly ($\beta = 100$, base currents included), once with $|V_{BE}| = 0.7\,\mathrm{V}$ and once with each transistor's $|V_{BE}|$ changed to 0.6 and 0.8 V:

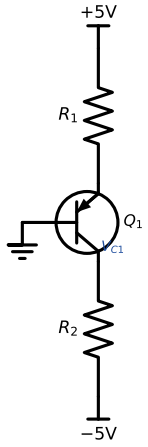
	target I_C	exact I_C	$\lvert V_{BE}\rvert = 0.6$	$\lvert V_{BE}\rvert = 0.8$	max change
Q_1	5 mA	4.83 mA	4.93 mA	4.74 mA	1.9 %

A $\pm 0.1\,\mathrm{V}$ change of V_{BE} moves no current by more than about 10 % of its design value (V_{BE}-insensitive), and the exact currents are close to the targets (I_B negligible).

Answer

$R_1 = 4.3\,\mathrm{k\Omega}, \quad R_2 = 5.7\,\mathrm{k\Omega}, \quad R_3 = 1\,\mathrm{k\Omega}$

Solution 4C.5 **Easy**



Step 1 — Rules

With $I_B \approx 0$ and every transistor active ($|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$), a resistor is known as soon as the voltages at both of its ends and the current through it are known: $R = \Delta V / I$. The specifications give the currents and some node voltages; the remaining freedom is removed with two rules:

- V_{BE}-insensitive: the voltage across the resistor that sets an emitter current must be at least 1 V (then $\pm 0.1\,\text{V}$ of V_{BE} changes I_C by at most 10 %);
- β-insensitive: a divider current must be much larger ($\geq 10\times$) than the base current it feeds, and a collector current that drives a base must be much larger than that base current.

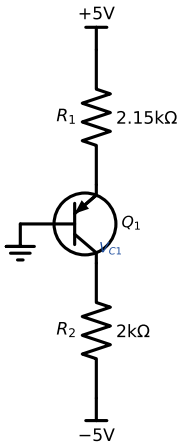
Work from the nodes that are already fixed towards the unknown ones.

Step 2 — Design, node by node

1. Q_1 is active: $V_{E1} = 0 + 0.7 = 0.7\,\text{V}$.
2. With $I_B \approx 0$ the only current through R_1 is I_{E1}: $I_{R1} = I_{E1} = 2\,\text{mA}$ ($I_C \approx I_E$).
3. $R_1 = \dfrac{5 - V_{E1}}{I_{E1}} = \dfrac{5 - 0.7}{2\,\text{mA}} = \mathbf{2.15\,k\Omega}$
4. With $I_B \approx 0$ the only current through R_2 is I_{C1}: $I_{R2} = I_{C1} = 2\,\text{mA}$.
5. $R_2 = \dfrac{V_{C1} - (-5)}{I_{C1}} = \dfrac{-1 - (-5)}{2\,\text{mA}} = \mathbf{2\,k\Omega}$

Step 3 — Designed circuit

resistor	value
R_1	$2.15\,\text{k}\Omega$
R_2	$2\,\text{k}\Omega$



Step 4 — Check the design

- $V_{EC1} = V_{E1} - V_{C1} = 0.7 - (-1) = 1.7\,\text{V} \geq 0.2\,\text{V}$ ✓

Finally the designed circuit is solved exactly ($\beta = 100$, base currents included), once with $|V_{BE}| = 0.7\,\text{V}$ and once with each transistor's $|V_{BE}|$ changed to 0.6 and 0.8 V:

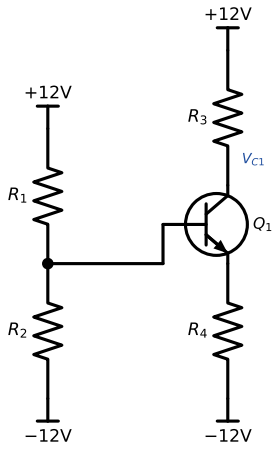
	target I_C	exact I_C	$\lvert V_{BE}\rvert = 0.6$	$\lvert V_{BE}\rvert = 0.8$	max change
Q_1	2 mA	1.98 mA	2.03 mA	1.93 mA	2.3 %

A $\pm 0.1\,\text{V}$ change of V_{BE} moves no current by more than about 10 % of its design value (V_{BE}-insensitive), and the exact currents are close to the targets (I_B negligible).

Answer

$R_1 = 2.15\,\text{k}\Omega, \quad R_2 = 2\,\text{k}\Omega$

Solution 4C.6 **Easy**



Step 1 — Rules

With $I_B \approx 0$ and every transistor active ($|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$), a resistor is known as soon as the voltages at both of its ends and the current through it are known: $R = \Delta V / I$. The specifications give the currents and some node voltages; the remaining freedom is removed with two rules:

- V_{BE}-insensitive: the voltage across the resistor that sets an emitter current must be at least 1 V (then $\pm 0.1\,\text{V}$ of V_{BE} changes I_C by at most 10 %);
- β-insensitive: a divider current must be much larger ($\geq 10\times$) than the base current it feeds, and a collector current that drives a base must be much larger than that base current.

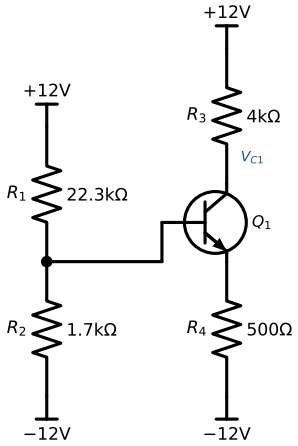
Work from the nodes that are already fixed towards the unknown ones.

Step 2 — Design, node by node

1. V_{BE}-insensitivity: a 0.1 V change of V_{BE} changes the voltage across R_4 by 0.1 V, so $\frac{\Delta I_C}{I_C} = \frac{0.1}{V_{R4}} \leq 10\%$ needs $V_{R4} \geq 1\,\text{V}$. Choose the smallest value, $V_{R4} = 1\,\text{V}$: $V_{E1} = (-12) + V_{R4} = -12 + 1 = -11\,\text{V}$.
2. The divider current must be much larger than the base current it feeds ($I_{B1} = 20\,\mu\text{A}$). Choose $I_{R2} = 1\,\text{mA}$ (50 times larger).
3. Q_1 is active: $V_{B1} = V_{E1} + 0.7 = -10.3\,\text{V}$.
4. With $I_B \approx 0$ the only current through R_1 is I_{R2}: $I_{R1} = I_{R2} = 1\,\text{mA}$.
5. $R_1 = \dfrac{12 - V_{B1}}{I_{R2}} = \dfrac{12 - (-10.3)}{1\,\text{mA}} = \mathbf{22.3\,k\Omega}$
6. $R_2 = \dfrac{V_{B1} - (-12)}{I_{R2}} = \dfrac{-10.3 - (-12)}{1\,\text{mA}} = \mathbf{1.7\,k\Omega}$
7. With $I_B \approx 0$ the only current through R_3 is I_{C1}: $I_{R3} = I_{C1} = 2\,\text{mA}$.
8. $R_3 = \dfrac{12 - V_{C1}}{I_{C1}} = \dfrac{12 - 4}{2\,\text{mA}} = \mathbf{4\,k\Omega}$
9. With $I_B \approx 0$ the only current through R_4 is I_{E1}: $I_{R4} = I_{E1} = 2\,\text{mA}$ ($I_C \approx I_E$).
10. $R_4 = \dfrac{V_{E1} - (-12)}{I_{E1}} = \dfrac{-11 - (-12)}{2\,\text{mA}} = \mathbf{500\,\Omega}$

Step 3 — Designed circuit

resistor	value
R_1	$22.3\,\mathrm{k\Omega}$
R_2	$1.7\,\mathrm{k\Omega}$
R_3	$4\,\mathrm{k\Omega}$
R_4	$500\,\Omega$



Step 4 — Check the design

- $I_{B1} = \frac{I_{C1}}{\beta} = 20\,\mu\mathrm{A} \ll$ the 1 mA flowing in the base network ✓
- $V_{CE1} = V_{C1} - V_{E1} = 4 - (-11) = 15\,\mathrm{V} \geq 0.2\,\mathrm{V}$ ✓

Finally the designed circuit is solved exactly ($\beta = 100$, base currents included), once with $|V_{BE}| = 0.7\,\mathrm{V}$ and once with each transistor's $|V_{BE}|$ changed to 0.6 and 0.8 V:

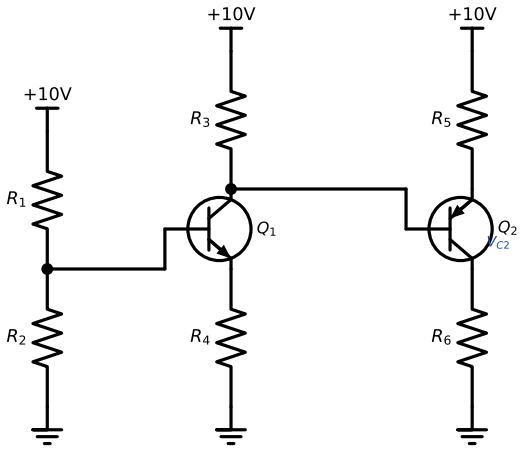
	target I_C	exact I_C	$\lvert V_{BE}\rvert = 0.6$	$\lvert V_{BE}\rvert = 0.8$	max change
Q_1	2 mA	1.92 mA	2.11 mA	1.73 mA	9.6 %

A $\pm 0.1\,\mathrm{V}$ change of V_{BE} moves no current by more than about 10 % of its design value (V_{BE}-insensitive), and the exact currents are close to the targets (I_B negligible).

Answer

$R_1 = 22.3\,\mathrm{k\Omega}, \quad R_2 = 1.7\,\mathrm{k\Omega}, \quad R_3 = 4\,\mathrm{k\Omega}, \quad R_4 = 500\,\Omega$

Solution 4C.7 **Medium**



Step 1 — Rules

With $I_B \approx 0$ and every transistor active ($|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$), a resistor is known as soon as the voltages at both of its ends and the current through it are known: $R = \Delta V / I$. The specifications give the currents and some node voltages; the remaining freedom is removed with two rules:

- V_{BE}-insensitive: the voltage across the resistor that sets an emitter current must be at least 1 V (then $\pm 0.1\,\text{V}$ of V_{BE} changes I_C by at most 10 %);
- β-insensitive: a divider current must be much larger ($\geq 10\times$) than the base current it feeds, and a collector current that drives a base must be much larger than that base current.

Work from the nodes that are already fixed towards the unknown ones.

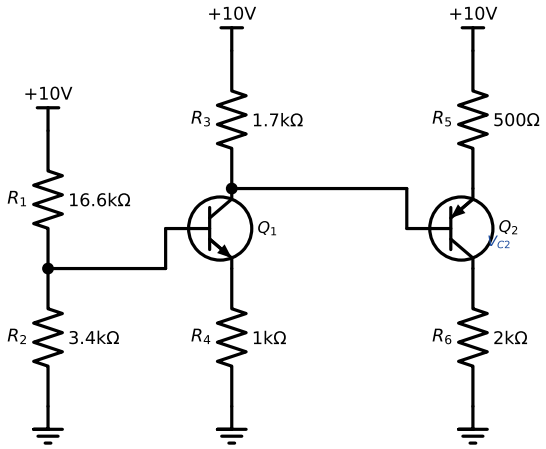
Step 2 — Design, node by node

1. V_{BE}-insensitivity: a 0.1 V change of V_{BE} changes the voltage across R_4 by 0.1 V, so $\frac{\Delta I_C}{I_C} = \frac{0.1}{V_{R4}} \leq 10\%$ needs $V_{R4} \geq 1\,\text{V}$. Choose the smallest value, $V_{R4} = 1\,\text{V}$: $V_{E1} = 0 + V_{R4} = 0 + 1 = 1\,\text{V}$.
2. V_{BE}-insensitivity: a 0.1 V change of V_{BE} changes the voltage across R_5 by 0.1 V, so $\frac{\Delta I_C}{I_C} = \frac{0.1}{V_{R5}} \leq 10\%$ needs $V_{R5} \geq 1\,\text{V}$. Choose the smallest value, $V_{R5} = 1\,\text{V}$: $V_{E2} = 10 - V_{R5} = 10 - 1 = 9\,\text{V}$.
3. The divider current must be much larger than the base current it feeds ($I_{B1} = 10\,\mu\text{A}$). Choose $I_{R2} = 500\,\mu\text{A}$ (50 times larger).
4. Q_1 is active: $V_{B1} = V_{E1} + 0.7 = 1.7\,\text{V}$.
5. Q_2 is active: $V_{C1} = V_{E2} - 0.7 = 8.3\,\text{V}$.
6. With $I_B \approx 0$ the only current through R_1 is I_{R2}: $I_{R1} = I_{R2} = 500\,\mu\text{A}$.
7. $R_1 = \dfrac{10 - V_{B1}}{I_{R2}} = \dfrac{10 - 1.7}{500\,\mu\text{A}} = \mathbf{16.6\,k\Omega}$
8. $R_2 = \dfrac{V_{B1} - 0}{I_{R2}} = \dfrac{1.7 - 0}{500\,\mu\text{A}} = \mathbf{3.4\,k\Omega}$
9. With $I_B \approx 0$ the only current through R_3 is I_{C1}: $I_{R3} = I_{C1} = 1\,\text{mA}$.
10. $R_3 = \dfrac{10 - V_{C1}}{I_{C1}} = \dfrac{10 - 8.3}{1\,\text{mA}} = \mathbf{1.7\,k\Omega}$

11. With $I_B \approx 0$ the only current through R_4 is I_{E1}: $I_{R4} = I_{E1} = 1\,\text{mA}$ ($I_C \approx I_E$).
12. $R_4 = \dfrac{V_{E1} - 0}{I_{E1}} = \dfrac{1 - 0}{1\,\text{mA}} = \mathbf{1\,k\Omega}$
13. With $I_B \approx 0$ the only current through R_5 is I_{E2}: $I_{R5} = I_{E2} = 2\,\text{mA}$ ($I_C \approx I_E$).
14. $R_5 = \dfrac{10 - V_{E2}}{I_{E2}} = \dfrac{10 - 9}{2\,\text{mA}} = \mathbf{500\,\Omega}$
15. With $I_B \approx 0$ the only current through R_6 is I_{C2}: $I_{R6} = I_{C2} = 2\,\text{mA}$.
16. $R_6 = \dfrac{V_{C2} - 0}{I_{C2}} = \dfrac{4 - 0}{2\,\text{mA}} = \mathbf{2\,k\Omega}$

Step 3 — Designed circuit

resistor	value
R_1	$16.6\,\text{k}\Omega$
R_2	$3.4\,\text{k}\Omega$
R_3	$1.7\,\text{k}\Omega$
R_4	$1\,\text{k}\Omega$
R_5	$500\,\Omega$
R_6	$2\,\text{k}\Omega$



Step 4 — Check the design

- $I_{B1} = \frac{I_{C1}}{\beta} = 10\,\mu\text{A} \ll$ the $500\,\mu\text{A}$ flowing in the base network ✓
- $I_{B2} = \frac{I_{C2}}{\beta} = 20\,\mu\text{A} \ll$ the $1\,\text{mA}$ flowing in the base network ✓
- $V_{CE1} = V_{C1} - V_{E1} = 8.3 - 1 = 7.3\,\text{V} \geq 1\,\text{V}$ ✓
- $V_{EC2} = V_{E2} - V_{C2} = 9 - 4 = 5\,\text{V} \geq 1\,\text{V}$ ✓

Finally the designed circuit is solved exactly ($\beta = 100$, base currents included), once with $|V_{BE}| = 0.7\,\text{V}$ and once with each transistor's $|V_{BE}|$ changed to 0.6 and $0.8\,\text{V}$:

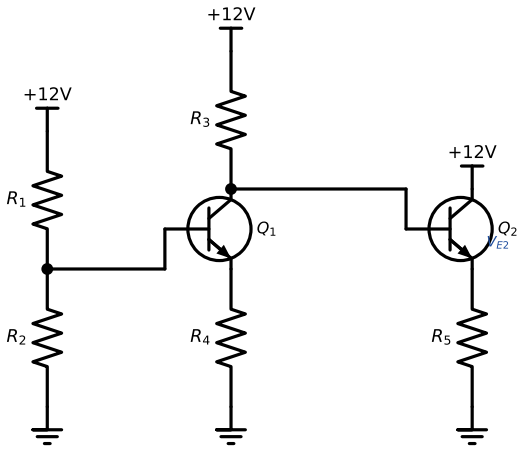
	target I_C	exact I_C	$\lvert V_{BE}\rvert = 0.6$	$\lvert V_{BE}\rvert = 0.8$	max change
Q_1	1 mA	963 μA	1.06 mA	867 μA	9.6 %
Q_2	2 mA	1.8 mA	1.99 mA	1.6 mA	9.6 %

A ± 0.1 V change of V_{BE} moves no current by more than about 10 % of its design value (V_{BE}-insensitive), and the exact currents are close to the targets (I_B negligible). The exact currents drift further from the targets along a chain of direct-coupled stages: each base current and each $I_C = \alpha I_E$ slightly shifts the collector voltage that biases the next stage, and with only ≈ 1 V across the next emitter resistor that shift is amplified. This is the normal accuracy of a hand design with $I_B \approx 0$; a larger emitter-resistor voltage in the later stages reduces the drift.

Answer

$R_1 = 16.6\,\text{k}\Omega, \quad R_2 = 3.4\,\text{k}\Omega, \quad R_3 = 1.7\,\text{k}\Omega, \quad R_4 = 1\,\text{k}\Omega, \quad R_5 = 500\,\Omega, \quad R_6 = 2\,\text{k}\Omega$

Solution 4C.8 **Medium**



Step 1 — Rules

With $I_B \approx 0$ and every transistor active ($|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$), a resistor is known as soon as the voltages at both of its ends and the current through it are known: $R = \Delta V / I$. The specifications give the currents and some node voltages; the remaining freedom is removed with two rules:

- V_{BE}-insensitive: the voltage across the resistor that sets an emitter current must be at least 1 V (then $\pm 0.1\,\text{V}$ of V_{BE} changes I_C by at most 10 %);
- β-insensitive: a divider current must be much larger ($\geq 10\times$) than the base current it feeds, and a collector current that drives a base must be much larger than that base current.

Work from the nodes that are already fixed towards the unknown ones.

Step 2 — Design, node by node

1. V_{BE}-insensitivity: a 0.1 V change of V_{BE} changes the voltage across R_4 by 0.1 V, so $\frac{\Delta I_C}{I_C} = \frac{0.1}{V_{R4}} \leq 10\%$ needs $V_{R4} \geq 1\,\text{V}$. Choose the smallest value, $V_{R4} = 1\,\text{V}$: $V_{E1} = 0 + V_{R4} = 0 + 1 = 1\,\text{V}$.
2. The divider current must be much larger than the base current it feeds ($I_{B1} = 10\,\mu\text{A}$). Choose $I_{R2} = 500\,\mu\text{A}$ (50 times larger).
3. Q_1 is active: $V_{B1} = V_{E1} + 0.7 = 1.7\,\text{V}$.
4. Q_2 is active: $V_{C1} = V_{E2} + 0.7 = 6.7\,\text{V}$.
5. With $I_B \approx 0$ the only current through R_1 is I_{R2}: $I_{R1} = I_{R2} = 500\,\mu\text{A}$.
6. $R_1 = \dfrac{12 - V_{B1}}{I_{R2}} = \dfrac{12 - 1.7}{500\,\mu\text{A}} = \mathbf{20.6\,k\Omega}$
7. $R_2 = \dfrac{V_{B1} - 0}{I_{R2}} = \dfrac{1.7 - 0}{500\,\mu\text{A}} = \mathbf{3.4\,k\Omega}$
8. With $I_B \approx 0$ the only current through R_3 is I_{C1}: $I_{R3} = I_{C1} = 1\,\text{mA}$.
9. $R_3 = \dfrac{12 - V_{C1}}{I_{C1}} = \dfrac{12 - 6.7}{1\,\text{mA}} = \mathbf{5.3\,k\Omega}$
10. With $I_B \approx 0$ the only current through R_4 is I_{E1}: $I_{R4} = I_{E1} = 1\,\text{mA}$ ($I_C \approx I_E$).

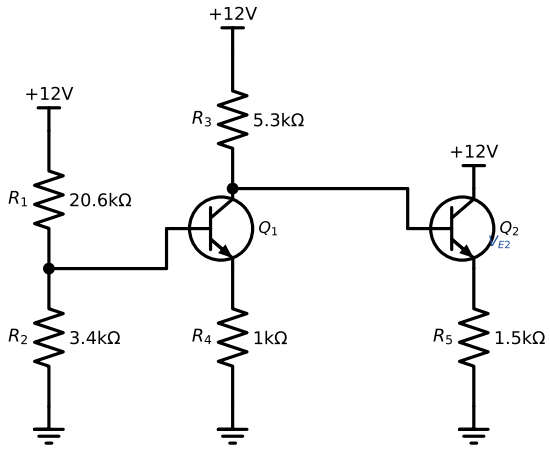
11. $R_4 = \dfrac{V_{E1} - 0}{I_{E1}} = \dfrac{1 - 0}{1\,\text{mA}} = \mathbf{1\,k\Omega}$

12. With $I_B \approx 0$ the only current through R_5 is I_{E2}: $I_{R5} = I_{E2} = 4\,\text{mA}$ ($I_C \approx I_E$).

13. $R_5 = \dfrac{V_{E2} - 0}{I_{E2}} = \dfrac{6 - 0}{4\,\text{mA}} = \mathbf{1.5\,k\Omega}$

Step 3 — Designed circuit

resistor	value
R_1	20.6 kΩ
R_2	3.4 kΩ
R_3	5.3 kΩ
R_4	1 kΩ
R_5	1.5 kΩ



Step 4 — Check the design

- $I_{B1} = \frac{I_{C1}}{\beta} = 10\,\mu\text{A} \ll$ the 500 μA flowing in the base network ✓
- $I_{B2} = \frac{I_{C2}}{\beta} = 40\,\mu\text{A} \ll$ the 1 mA flowing in the base network ✓
- $V_{CE1} = V_{C1} - V_{E1} = 6.7 - 1 = 5.7\,\text{V} \geq 1\,\text{V}$ ✓
- $V_{CE2} = 12 - V_{E2} = 12 - 6 = 6\,\text{V} \geq 1\,\text{V}$ ✓

Finally the designed circuit is solved exactly ($\beta = 100$, base currents included), once with $|V_{BE}| = 0.7\,\text{V}$ and once with each transistor's $|V_{BE}|$ changed to 0.6 and 0.8 V:

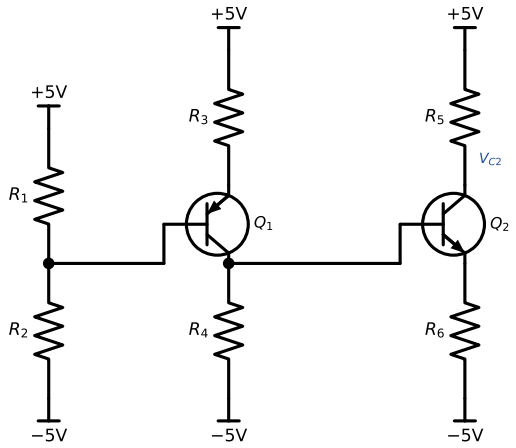
	target I_C	exact I_C	$\lvert V_{BE}\rvert = 0.6$	$\lvert V_{BE}\rvert = 0.8$	max change
Q_1	1 mA	962 μA	1.06 mA	866 μA	9.6 %
Q_2	4 mA	3.95 mA	4.02 mA	3.89 mA	1.6 %

A $\pm 0.1\,\mathrm{V}$ change of V_{BE} moves no current by more than about $10\,\%$ of its design value (V_{BE}-insensitive), and the exact currents are close to the targets (I_B negligible).

Answer

$R_1 = 20.6\,\mathrm{k\Omega}, \quad R_2 = 3.4\,\mathrm{k\Omega}, \quad R_3 = 5.3\,\mathrm{k\Omega}, \quad R_4 = 1\,\mathrm{k\Omega}, \quad R_5 = 1.5\,\mathrm{k\Omega}$

Solution 4C.9 **Medium**



Step 1 — Rules

With $I_B \approx 0$ and every transistor active ($|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$), a resistor is known as soon as the voltages at both of its ends and the current through it are known: $R = \Delta V / I$. The specifications give the currents and some node voltages; the remaining freedom is removed with two rules:

- V_{BE}-insensitive: the voltage across the resistor that sets an emitter current must be at least 1 V (then $\pm 0.1\,\text{V}$ of V_{BE} changes I_C by at most 10 %);
- β-insensitive: a divider current must be much larger ($\geq 10\times$) than the base current it feeds, and a collector current that drives a base must be much larger than that base current.

Work from the nodes that are already fixed towards the unknown ones.

Step 2 — Design, node by node

1. V_{BE}-insensitivity: a 0.1 V change of V_{BE} changes the voltage across R_3 by 0.1 V, so $\frac{\Delta I_C}{I_C} = \frac{0.1}{V_{R3}} \leq 10\%$ needs $V_{R3} \geq 1\,\text{V}$. Choose the smallest value, $V_{R3} = 1\,\text{V}$: $V_{E1} = 5 - V_{R3} = 5 - 1 = 4\,\text{V}$.
2. V_{BE}-insensitivity: a 0.1 V change of V_{BE} changes the voltage across R_6 by 0.1 V, so $\frac{\Delta I_C}{I_C} = \frac{0.1}{V_{R6}} \leq 10\%$ needs $V_{R6} \geq 1\,\text{V}$. Choose the smallest value, $V_{R6} = 1\,\text{V}$: $V_{E2} = (-5) + V_{R6} = -5 + 1 = -4\,\text{V}$.
3. The divider current must be much larger than the base current it feeds ($I_{B1} = 10\,\mu\text{A}$). Choose $I_{R2} = 500\,\mu\text{A}$ (50 times larger).
4. Q_1 is active: $V_{B1} = V_{E1} - 0.7 = 3.3\,\text{V}$.
5. Q_2 is active: $V_{C1} = V_{E2} + 0.7 = -3.3\,\text{V}$.
6. With $I_B \approx 0$ the only current through R_1 is I_{R2}: $I_{R1} = I_{R2} = 500\,\mu\text{A}$.
7. $R_1 = \dfrac{5 - V_{B1}}{I_{R2}} = \dfrac{5 - 3.3}{500\,\mu\text{A}} = \mathbf{3.4\,k\Omega}$
8. $R_2 = \dfrac{V_{B1} - (-5)}{I_{R2}} = \dfrac{3.3 - (-5)}{500\,\mu\text{A}} = \mathbf{16.6\,k\Omega}$
9. With $I_B \approx 0$ the only current through R_3 is I_{E1}: $I_{R3} = I_{E1} = 1\,\text{mA}$ ($I_C \approx I_E$).
10. $R_3 = \dfrac{5 - V_{E1}}{I_{E1}} = \dfrac{5 - 4}{1\,\text{mA}} = \mathbf{1\,k\Omega}$

11. With $I_B \approx 0$ the only current through R_4 is I_{C1}: $I_{R4} = I_{C1} = 1\,\text{mA}$.

12. $R_4 = \dfrac{V_{C1} - (-5)}{I_{C1}} = \dfrac{-3.3 - (-5)}{1\,\text{mA}} = \mathbf{1.7\,k\Omega}$

13. With $I_B \approx 0$ the only current through R_5 is I_{C2}: $I_{R5} = I_{C2} = 2\,\text{mA}$.

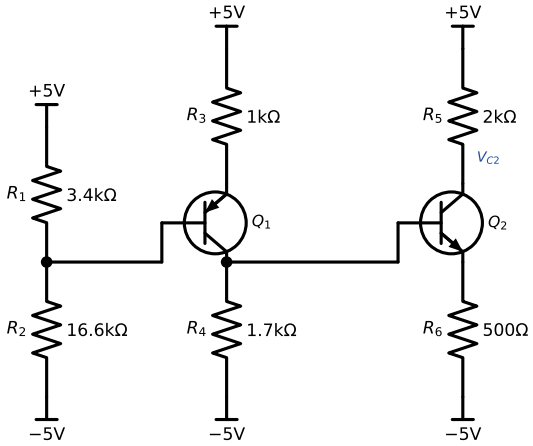
14. $R_5 = \dfrac{5 - V_{C2}}{I_{C2}} = \dfrac{5 - 1}{2\,\text{mA}} = \mathbf{2\,k\Omega}$

15. With $I_B \approx 0$ the only current through R_6 is I_{E2}: $I_{R6} = I_{E2} = 2\,\text{mA}$ ($I_C \approx I_E$).

16. $R_6 = \dfrac{V_{E2} - (-5)}{I_{E2}} = \dfrac{-4 - (-5)}{2\,\text{mA}} = \mathbf{500\,\Omega}$

Step 3 — Designed circuit

resistor	value
R_1	$3.4\,\text{k}\Omega$
R_2	$16.6\,\text{k}\Omega$
R_3	$1\,\text{k}\Omega$
R_4	$1.7\,\text{k}\Omega$
R_5	$2\,\text{k}\Omega$
R_6	$500\,\Omega$



Step 4 — Check the design

- $I_{B1} = \frac{I_{C1}}{\beta} = 10\,\mu\text{A} \ll$ the $500\,\mu\text{A}$ flowing in the base network ✓
- $I_{B2} = \frac{I_{C2}}{\beta} = 20\,\mu\text{A} \ll$ the $1\,\text{mA}$ flowing in the base network ✓
- $V_{EC1} = V_{E1} - V_{C1} = 4 - (-3.3) = 7.3\,\text{V} \geq 1\,\text{V}$ ✓
- $V_{CE2} = V_{C2} - V_{E2} = 1 - (-4) = 5\,\text{V} \geq 1\,\text{V}$ ✓

Finally the designed circuit is solved exactly ($\beta = 100$, base currents included), once with $|V_{BE}| = 0.7\,\text{V}$ and once with each transistor's $|V_{BE}|$ changed to 0.6 and $0.8\,\text{V}$:

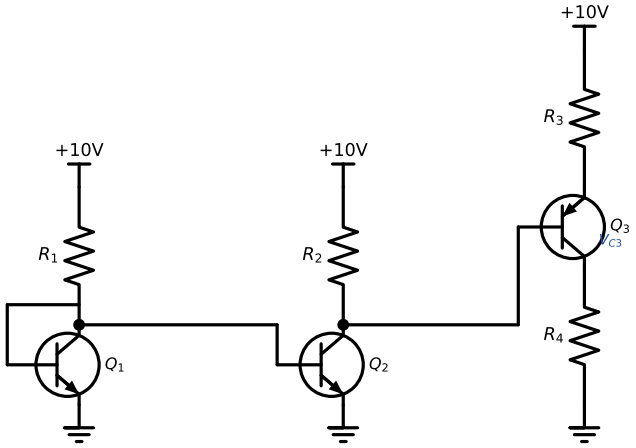
	target I_C	exact I_C	$\lvert V_{BE}\rvert = 0.6$	$\lvert V_{BE}\rvert = 0.8$	max change
Q_1	1 mA	963 μA	1.06 mA	867 μA	9.6 %
Q_2	2 mA	1.8 mA	1.99 mA	1.6 mA	9.6 %

A ± 0.1 V change of V_{BE} moves no current by more than about 10 % of its design value (V_{BE}-insensitive), and the exact currents are close to the targets (I_B negligible). The exact currents drift further from the targets along a chain of direct-coupled stages: each base current and each $I_C = \alpha I_E$ slightly shifts the collector voltage that biases the next stage, and with only ≈ 1 V across the next emitter resistor that shift is amplified. This is the normal accuracy of a hand design with $I_B \approx 0$; a larger emitter-resistor voltage in the later stages reduces the drift.

Answer

$R_1 = 3.4\,\text{k}\Omega$, $R_2 = 16.6\,\text{k}\Omega$, $R_3 = 1\,\text{k}\Omega$, $R_4 = 1.7\,\text{k}\Omega$, $R_5 = 2\,\text{k}\Omega$, $R_6 = 500\,\Omega$

Solution 4C.10 **Medium**



Step 1 — Rules

With $I_B \approx 0$ and every transistor active ($|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$), a resistor is known as soon as the voltages at both of its ends and the current through it are known: $R = \Delta V / I$. The specifications give the currents and some node voltages; the remaining freedom is removed with two rules:

- V_{BE}-insensitive: the voltage across the resistor that sets an emitter current must be at least 1 V (then $\pm 0.1\,\text{V}$ of V_{BE} changes I_C by at most 10 %);
- β-insensitive: a divider current must be much larger ($\geq 10\times$) than the base current it feeds, and a collector current that drives a base must be much larger than that base current.

Work from the nodes that are already fixed towards the unknown ones.

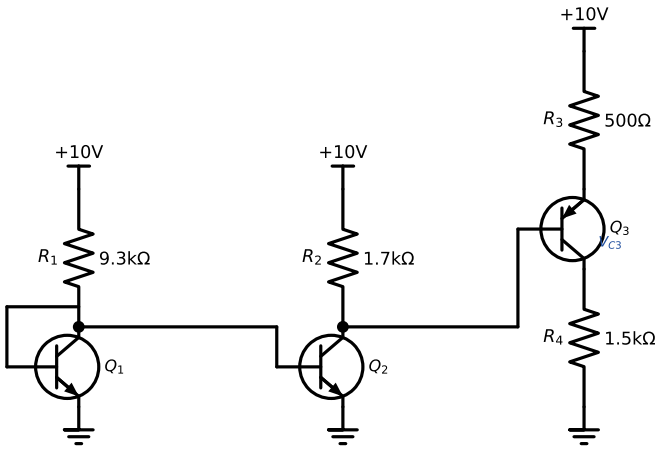
Step 2 — Design, node by node

1. V_{BE}-insensitivity: a 0.1 V change of V_{BE} changes the voltage across R_3 by 0.1 V, so $\frac{\Delta I_C}{I_C} = \frac{0.1}{V_{R3}} \leq 10\%$ needs $V_{R3} \geq 1\,\text{V}$. Choose the smallest value, $V_{R3} = 1\,\text{V}$: $V_{E3} = 10 - V_{R3} = 10 - 1 = 9\,\text{V}$.
2. Q_1 (diode-connected) is active: $V_{B1} = 0 + 0.7 = 0.7\,\text{V}$.
3. Q_3 is active: $V_{C2} = V_{E3} - 0.7 = 8.3\,\text{V}$.
4. Q_2 mirrors Q_1 (same base and emitter nodes): $I_{C2} = I_{C1} = 1\,\text{mA}$.
5. With $I_B \approx 0$ the only current through R_1 is I_{C1}: $I_{R1} = I_{C1} = 1\,\text{mA}$.
6. $R_1 = \dfrac{10 - V_{B1}}{I_{C1}} = \dfrac{10 - 0.7}{1\,\text{mA}} = \mathbf{9.3\,k\Omega}$
7. With $I_B \approx 0$ the only current through R_2 is I_{C2}: $I_{R2} = I_{C2} = 1\,\text{mA}$.
8. $R_2 = \dfrac{10 - V_{C2}}{I_{C2}} = \dfrac{10 - 8.3}{1\,\text{mA}} = \mathbf{1.7\,k\Omega}$
9. With $I_B \approx 0$ the only current through R_3 is I_{E3}: $I_{R3} = I_{E3} = 2\,\text{mA}$ ($I_C \approx I_E$).
10. $R_3 = \dfrac{10 - V_{E3}}{I_{E3}} = \dfrac{10 - 9}{2\,\text{mA}} = \mathbf{500\,\Omega}$
11. With $I_B \approx 0$ the only current through R_4 is I_{C3}: $I_{R4} = I_{C3} = 2\,\text{mA}$.

12. $R_4 = \dfrac{V_{C3} - 0}{I_{C3}} = \dfrac{3 - 0}{2\,\text{mA}} = \mathbf{1.5\,k\Omega}$

Step 3 — Designed circuit

resistor	value
R_1	$9.3\,\text{k}\Omega$
R_2	$1.7\,\text{k}\Omega$
R_3	$500\,\Omega$
R_4	$1.5\,\text{k}\Omega$



Step 4 — Check the design

- $I_{B1} = \frac{I_{C1}}{\beta} = 10\,\mu\text{A}$ (all bases on this node: $20\,\mu\text{A}$) $\ll$ the $1\,\text{mA}$ flowing in the base network ✓
- $I_{B2} = \frac{I_{C2}}{\beta} = 10\,\mu\text{A}$ (all bases on this node: $20\,\mu\text{A}$) $\ll$ the $1\,\text{mA}$ flowing in the base network ✓
- $I_{B3} = \frac{I_{C3}}{\beta} = 20\,\mu\text{A} \ll$ the $1\,\text{mA}$ flowing in the base network ✓
- Q_1 is diode-connected: $V_{CE1} = |V_{BE}| = 0.7\,\text{V}$, always active (the $|V_{CE}|$ specification is meant for the other transistors) ✓
- $V_{CE2} = V_{C2} - 0 = 8.3 - 0 = 8.3\,\text{V} \geq 1\,\text{V}$ ✓
- $V_{EC3} = V_{E3} - V_{C3} = 9 - 3 = 6\,\text{V} \geq 1\,\text{V}$ ✓

Finally the designed circuit is solved exactly ($\beta = 100$, base currents included), once with $|V_{BE}| = 0.7\,\text{V}$ and once with each transistor's $|V_{BE}|$ changed to 0.6 and 0.8 V:

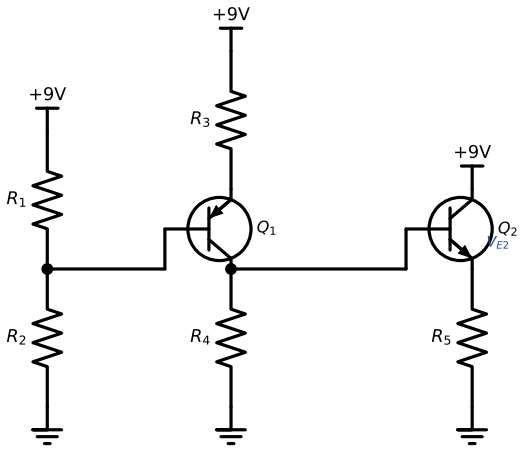
	target I_C	exact I_C	$\lvert V_{BE}\rvert = 0.6$	$\lvert V_{BE}\rvert = 0.8$	max change
Q_1	$1\,\text{mA}$	$980\,\mu\text{A}$	$991\,\mu\text{A}$	$970\,\mu\text{A}$	$1.1\,\%$
Q_2	$1\,\text{mA}$	$980\,\mu\text{A}$	$991\,\mu\text{A}$	$970\,\mu\text{A}$	$1.1\,\%$
Q_3	$2\,\text{mA}$	$1.85\,\text{mA}$	$2.04\,\text{mA}$	$1.66\,\text{mA}$	$9.6\,\%$

A $\pm 0.1\,\mathrm{V}$ change of V_{BE} moves no current by more than about $10\,\%$ of its design value (V_{BE}-insensitive), and the exact currents are close to the targets (I_B negligible). The exact currents drift further from the targets along a chain of direct-coupled stages: each base current and each $I_C = \alpha I_E$ slightly shifts the collector voltage that biases the next stage, and with only $\approx 1\,\mathrm{V}$ across the next emitter resistor that shift is amplified. This is the normal accuracy of a hand design with $I_B \approx 0$; a larger emitter-resistor voltage in the later stages reduces the drift.

Answer

$R_1 = 9.3\,\mathrm{k\Omega}, \quad R_2 = 1.7\,\mathrm{k\Omega}, \quad R_3 = 500\,\Omega, \quad R_4 = 1.5\,\mathrm{k\Omega}$

Solution 4C.11 **Medium**



Step 1 — Rules

With $I_B \approx 0$ and every transistor active ($|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$), a resistor is known as soon as the voltages at both of its ends and the current through it are known: $R = \Delta V / I$. The specifications give the currents and some node voltages; the remaining freedom is removed with two rules:

- V_{BE}-insensitive: the voltage across the resistor that sets an emitter current must be at least 1 V (then $\pm 0.1\,\text{V}$ of V_{BE} changes I_C by at most 10 %);
- β-insensitive: a divider current must be much larger ($\geq 10\times$) than the base current it feeds, and a collector current that drives a base must be much larger than that base current.

Work from the nodes that are already fixed towards the unknown ones.

Step 2 — Design, node by node

1. V_{BE}-insensitivity: a 0.1 V change of V_{BE} changes the voltage across R_3 by 0.1 V, so $\frac{\Delta I_C}{I_C} = \frac{0.1}{V_{R3}} \leq 10\%$ needs $V_{R3} \geq 1\,\text{V}$. Choose the smallest value, $V_{R3} = 1\,\text{V}$: $V_{E1} = 9 - V_{R3} = 9 - 1 = 8\,\text{V}$.
2. The divider current must be much larger than the base current it feeds ($I_{B1} = 10\,\mu\text{A}$). Choose $I_{R2} = 500\,\mu\text{A}$ (50 times larger).
3. Q_1 is active: $V_{B1} = V_{E1} - 0.7 = 7.3\,\text{V}$.
4. Q_2 is active: $V_{C1} = V_{E2} + 0.7 = 3.7\,\text{V}$.
5. With $I_B \approx 0$ the only current through R_1 is I_{R2}: $I_{R1} = I_{R2} = 500\,\mu\text{A}$.
6. $R_1 = \dfrac{9 - V_{B1}}{I_{R2}} = \dfrac{9 - 7.3}{500\,\mu\text{A}} = \mathbf{3.4\,k\Omega}$
7. $R_2 = \dfrac{V_{B1} - 0}{I_{R2}} = \dfrac{7.3 - 0}{500\,\mu\text{A}} = \mathbf{14.6\,k\Omega}$
8. With $I_B \approx 0$ the only current through R_3 is I_{E1}: $I_{R3} = I_{E1} = 1\,\text{mA}$ ($I_C \approx I_E$).
9. $R_3 = \dfrac{9 - V_{E1}}{I_{E1}} = \dfrac{9 - 8}{1\,\text{mA}} = \mathbf{1\,k\Omega}$
10. With $I_B \approx 0$ the only current through R_4 is I_{C1}: $I_{R4} = I_{C1} = 1\,\text{mA}$.

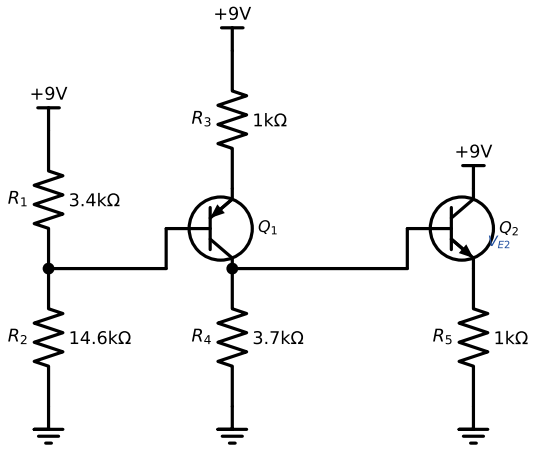
11. $R_4 = \dfrac{V_{C1} - 0}{I_{C1}} = \dfrac{3.7 - 0}{1\,\text{mA}} = \mathbf{3.7\,k\Omega}$

12. With $I_B \approx 0$ the only current through R_5 is I_{E2}: $I_{R5} = I_{E2} = 3\,\text{mA}$ ($I_C \approx I_E$).

13. $R_5 = \dfrac{V_{E2} - 0}{I_{E2}} = \dfrac{3 - 0}{3\,\text{mA}} = \mathbf{1\,k\Omega}$

Step 3 — Designed circuit

resistor	value
R_1	$3.4\,\text{k}\Omega$
R_2	$14.6\,\text{k}\Omega$
R_3	$1\,\text{k}\Omega$
R_4	$3.7\,\text{k}\Omega$
R_5	$1\,\text{k}\Omega$



Step 4 — Check the design

- $I_{B1} = \frac{I_{C1}}{\beta} = 10\,\mu\text{A} \ll$ the $500\,\mu\text{A}$ flowing in the base network ✓
- $I_{B2} = \frac{I_{C2}}{\beta} = 30\,\mu\text{A} \ll$ the $1\,\text{mA}$ flowing in the base network ✓
- $V_{EC1} = V_{E1} - V_{C1} = 8 - 3.7 = 4.3\,\text{V} \geq 1\,\text{V}$ ✓
- $V_{CE2} = 9 - V_{E2} = 9 - 3 = 6\,\text{V} \geq 1\,\text{V}$ ✓

Finally the designed circuit is solved exactly ($\beta = 100$, base currents included), once with $|V_{BE}| = 0.7\,\text{V}$ and once with each transistor's $|V_{BE}|$ changed to 0.6 and 0.8 V:

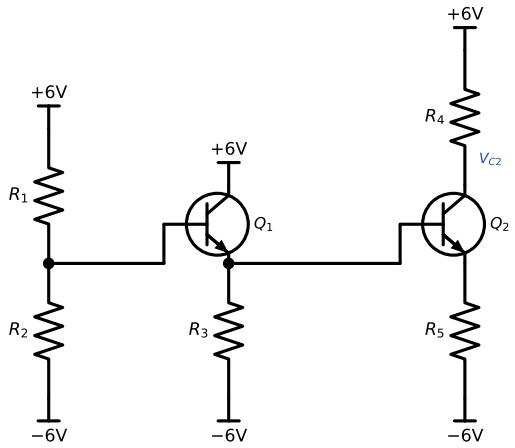
	target I_C	exact I_C	$\lvert V_{BE}\rvert = 0.6$	$\lvert V_{BE}\rvert = 0.8$	max change
Q_1	$1\,\text{mA}$	$964\,\mu\text{A}$	$1.06\,\text{mA}$	$867\,\mu\text{A}$	$9.6\,\%$
Q_2	$3\,\text{mA}$	$2.74\,\text{mA}$	$2.83\,\text{mA}$	$2.64\,\text{mA}$	$3.2\,\%$

A $\pm 0.1\,\mathrm{V}$ change of V_{BE} moves no current by more than about $10\,\%$ of its design value (V_{BE}-insensitive), and the exact currents are close to the targets (I_B negligible). The exact currents drift further from the targets along a chain of direct-coupled stages: each base current and each $I_C = \alpha I_E$ slightly shifts the collector voltage that biases the next stage, and with only $\approx 1\,\mathrm{V}$ across the next emitter resistor that shift is amplified. This is the normal accuracy of a hand design with $I_B \approx 0$; a larger emitter-resistor voltage in the later stages reduces the drift.

Answer

$R_1 = 3.4\,\mathrm{k\Omega}, \quad R_2 = 14.6\,\mathrm{k\Omega}, \quad R_3 = 1\,\mathrm{k\Omega}, \quad R_4 = 3.7\,\mathrm{k\Omega}, \quad R_5 = 1\,\mathrm{k\Omega}$

Solution 4C.12 **Medium**



Step 1 — Rules

With $I_B \approx 0$ and every transistor active ($|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$), a resistor is known as soon as the voltages at both of its ends and the current through it are known: $R = \Delta V/I$. The specifications give the currents and some node voltages; the remaining freedom is removed with two rules:

- V_{BE}-insensitive: the voltage across the resistor that sets an emitter current must be at least 1 V (then $\pm 0.1\,\text{V}$ of V_{BE} changes I_C by at most 10 %);
- β-insensitive: a divider current must be much larger ($\geq 10\times$) than the base current it feeds, and a collector current that drives a base must be much larger than that base current.

Work from the nodes that are already fixed towards the unknown ones.

Step 2 — Design, node by node

1. V_{BE}-insensitivity: a 0.1 V change of V_{BE} changes the voltage across R_5 by 0.1 V, so $\frac{\Delta I_C}{I_C} = \frac{0.1}{V_{R5}} \leq 10\%$ needs $V_{R5} \geq 1\,\text{V}$. Choose the smallest value, $V_{R5} = 1\,\text{V}$: $V_{E2} = (-6) + V_{R5} = -6 + 1 = -5\,\text{V}$.
2. The divider current must be much larger than the base current it feeds ($I_{B1} = 10\,\mu\text{A}$). Choose $I_{R2} = 500\,\mu\text{A}$ (50 times larger).
3. Q_2 is active: $V_{E1} = V_{E2} + 0.7 = -4.3\,\text{V}$.
4. With $I_B \approx 0$ the only current through R_1 is I_{R2}: $I_{R1} = I_{R2} = 500\,\mu\text{A}$.
5. Q_1 is active: $V_{B1} = V_{E1} + 0.7 = -3.6\,\text{V}$.
6. With $I_B \approx 0$ the only current through R_3 is I_{E1}: $I_{R3} = I_{E1} = 1\,\text{mA}$ ($I_C \approx I_E$).
7. $R_1 = \dfrac{6 - V_{B1}}{I_{R2}} = \dfrac{6 - (-3.6)}{500\,\mu\text{A}} = \mathbf{19.2\,k\Omega}$
8. $R_2 = \dfrac{V_{B1} - (-6)}{I_{R2}} = \dfrac{-3.6 - (-6)}{500\,\mu\text{A}} = \mathbf{4.8\,k\Omega}$
9. $R_3 = \dfrac{V_{E1} - (-6)}{I_{E1}} = \dfrac{-4.3 - (-6)}{1\,\text{mA}} = \mathbf{1.7\,k\Omega}$
10. With $I_B \approx 0$ the only current through R_4 is I_{C2}: $I_{R4} = I_{C2} = 2\,\text{mA}$.

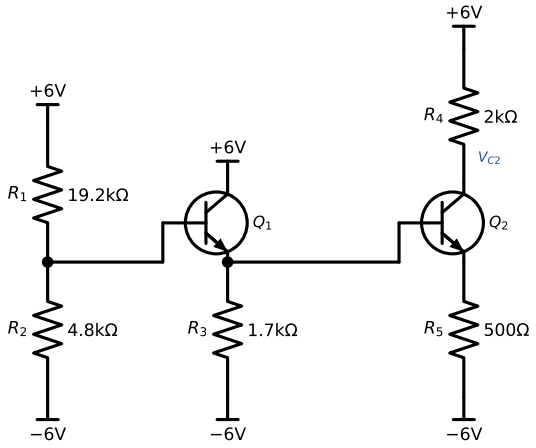
11. $R_4 = \dfrac{6 - V_{C2}}{I_{C2}} = \dfrac{6-2}{2\,\text{mA}} = \mathbf{2\,k\Omega}$

12. With $I_B \approx 0$ the only current through R_5 is I_{E2}: $I_{R5} = I_{E2} = 2\,\text{mA}$ ($I_C \approx I_E$).

13. $R_5 = \dfrac{V_{E2} - (-6)}{I_{E2}} = \dfrac{-5-(-6)}{2\,\text{mA}} = \mathbf{500\,\Omega}$

Step 3 — Designed circuit

resistor	value
R_1	$19.2\,\text{k}\Omega$
R_2	$4.8\,\text{k}\Omega$
R_3	$1.7\,\text{k}\Omega$
R_4	$2\,\text{k}\Omega$
R_5	$500\,\Omega$



Step 4 — Check the design

- $I_{B1} = \frac{I_{C1}}{\beta} = 10\,\mu\text{A} \ll$ the $500\,\mu\text{A}$ flowing in the base network ✓
- $I_{B2} = \frac{I_{C2}}{\beta} = 20\,\mu\text{A} \ll$ the $1\,\text{mA}$ flowing in the base network ✓
- $V_{CE1} = 6 - V_{E1} = 6 - (-4.3) = 10.3\,\text{V} \geq 1\,\text{V}$ ✓
- $V_{CE2} = V_{C2} - V_{E2} = 2 - (-5) = 7\,\text{V} \geq 1\,\text{V}$ ✓

Finally the designed circuit is solved exactly ($\beta = 100$, base currents included), once with $|V_{BE}| = 0.7\,\text{V}$ and once with each transistor's $|V_{BE}|$ changed to 0.6 and 0.8 V:

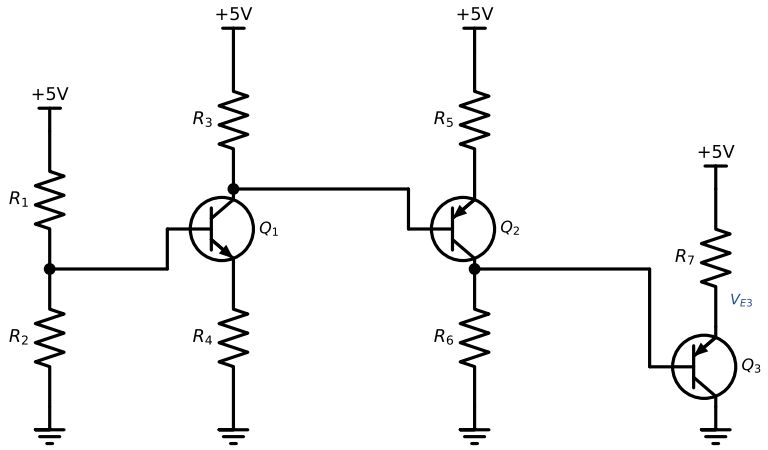
	target I_C	exact I_C	$\lvert V_{BE}\rvert = 0.6$	$\lvert V_{BE}\rvert = 0.8$	max change
Q_1	$1\,\text{mA}$	$987\,\mu\text{A}$	$1.05\,\text{mA}$	$928\,\mu\text{A}$	5.9 %
Q_2	$2\,\text{mA}$	$1.91\,\text{mA}$	$2.1\,\text{mA}$	$1.71\,\text{mA}$	9.9 %

A $\pm 0.1\,\mathrm{V}$ change of V_{BE} moves no current by more than about $10\,\%$ of its design value (V_{BE}-insensitive), and the exact currents are close to the targets (I_B negligible).

Answer

$R_1 = 19.2\,\mathrm{k\Omega}, \quad R_2 = 4.8\,\mathrm{k\Omega}, \quad R_3 = 1.7\,\mathrm{k\Omega}, \quad R_4 = 2\,\mathrm{k\Omega}, \quad R_5 = 500\,\Omega$

Solution 4C.13 **Hard**



Step 1 — Rules

With $I_B \approx 0$ and every transistor active ($|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$), a resistor is known as soon as the voltages at both of its ends and the current through it are known: $R = \Delta V / I$. The specifications give the currents and some node voltages; the remaining freedom is removed with two rules:

- V_{BE}-insensitive: the voltage across the resistor that sets an emitter current must be at least 1 V (then $\pm 0.1\,\text{V}$ of V_{BE} changes I_C by at most 10 %);
- β-insensitive: a divider current must be much larger ($\geq 10\times$) than the base current it feeds, and a collector current that drives a base must be much larger than that base current.

Work from the nodes that are already fixed towards the unknown ones.

Step 2 — Design, node by node

1. V_{BE}-insensitivity: a 0.1 V change of V_{BE} changes the voltage across R_5 by 0.1 V, so $\frac{\Delta I_C}{I_C} = \frac{0.1}{V_{R5}} \leq 10\%$ needs $V_{R5} \geq 1\,\text{V}$. Choose the smallest value, $V_{R5} = 1\,\text{V}$: $V_{E2} = 5 - V_{R5} = 5 - 1 = 4\,\text{V}$.
2. V_{BE}-insensitivity: a 0.1 V change of V_{BE} changes the voltage across R_4 by 0.1 V, so $\frac{\Delta I_C}{I_C} = \frac{0.1}{V_{R4}} \leq 10\%$ needs $V_{R4} \geq 1\,\text{V}$. Choose the smallest value, $V_{R4} = 1\,\text{V}$: $V_{E1} = 0 + V_{R4} = 0 + 1 = 1\,\text{V}$.
3. $I_{dc} = I_{R1} + I_{C1} + I_{C2} + I_{C3} \leq 14\,\text{mA}$ leaves at most $14 - 2 - 4 - 6 = 2\,\text{mA}$ for the divider, and I_{R2} must be $\gg I_{B1} = 20\,\mu\text{A}$. Choose $I_{R2} = 1\,\text{mA}$.
4. Q_1 is active: $V_{B1} = V_{E1} + 0.7 = 1.7\,\text{V}$.
5. Q_2 is active: $V_{C1} = V_{E2} - 0.7 = 3.3\,\text{V}$.
6. Q_3 is active: $V_{C2} = V_{E3} - 0.7 = 1.8\,\text{V}$.
7. With $I_B \approx 0$ the only current through R_1 is I_{R2}: $I_{R1} = I_{R2} = 1\,\text{mA}$.
8. $R_1 = \dfrac{5 - V_{B1}}{I_{R2}} = \dfrac{5 - 1.7}{1\,\text{mA}} = \mathbf{3.3\,k\Omega}$
9. $R_2 = \dfrac{V_{B1} - 0}{I_{R2}} = \dfrac{1.7 - 0}{1\,\text{mA}} = \mathbf{1.7\,k\Omega}$
10. With $I_B \approx 0$ the only current through R_3 is I_{C1}: $I_{R3} = I_{C1} = 2\,\text{mA}$.

11. $R_3 = \dfrac{5 - V_{C1}}{I_{C1}} = \dfrac{5 - 3.3}{2\,\text{mA}} = \mathbf{850\,\Omega}$

12. With $I_B \approx 0$ the only current through R_4 is I_{E1}: $I_{R4} = I_{E1} = 2\,\text{mA}$ ($I_C \approx I_E$).

13. $R_4 = \dfrac{V_{E1} - 0}{I_{E1}} = \dfrac{1 - 0}{2\,\text{mA}} = \mathbf{500\,\Omega}$

14. With $I_B \approx 0$ the only current through R_5 is I_{E2}: $I_{R5} = I_{E2} = 4\,\text{mA}$ ($I_C \approx I_E$).

15. $R_5 = \dfrac{5 - V_{E2}}{I_{E2}} = \dfrac{5 - 4}{4\,\text{mA}} = \mathbf{250\,\Omega}$

16. With $I_B \approx 0$ the only current through R_6 is I_{C2}: $I_{R6} = I_{C2} = 4\,\text{mA}$.

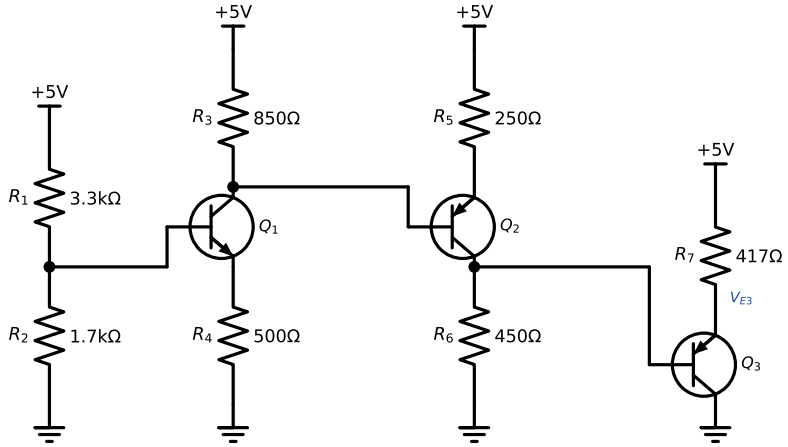
17. $R_6 = \dfrac{V_{C2} - 0}{I_{C2}} = \dfrac{1.8 - 0}{4\,\text{mA}} = \mathbf{450\,\Omega}$

18. With $I_B \approx 0$ the only current through R_7 is I_{E3}: $I_{R7} = I_{E3} = 6\,\text{mA}$ ($I_C \approx I_E$).

19. $R_7 = \dfrac{5 - V_{E3}}{I_{E3}} = \dfrac{5 - 2.5}{6\,\text{mA}} = \mathbf{417\,\Omega}$

Step 3 — Designed circuit

resistor	value
R_1	3.3 kΩ
R_2	1.7 kΩ
R_3	850 Ω
R_4	500 Ω
R_5	250 Ω
R_6	450 Ω
R_7	417 Ω



Step 4 — Check the design

- $I_{B1} = \frac{I_{C1}}{\beta} = 20\,\mu\text{A} \ll$ the 1 mA flowing in the base network ✓

- $I_{B2} = \frac{I_{C2}}{\beta} = 40\,\mu\text{A} \ll$ the 2 mA flowing in the base network ✓
- $I_{B3} = \frac{I_{C3}}{\beta} = 60\,\mu\text{A} \ll$ the 4 mA flowing in the base network ✓
- $V_{CE1} = V_{C1} - V_{E1} = 3.3 - 1 = 2.3\,\text{V} \geq 1\,\text{V}$ ✓
- $V_{EC2} = V_{E2} - V_{C2} = 4 - 1.8 = 2.2\,\text{V} \geq 1\,\text{V}$ ✓
- $V_{EC3} = V_{E3} - 0 = 2.5 - 0 = 2.5\,\text{V} \geq 1\,\text{V}$ ✓
- $I_{dc} = I_{R1} + I_{R3} + I_{R5} + I_{R7} = 1\,\text{mA} + 2\,\text{mA} + 4\,\text{mA} + 6\,\text{mA} = 13\,\text{mA} \leq 14\,\text{mA}$ ✓

Finally the designed circuit is solved exactly ($\beta = 100$, base currents included), once with $|V_{BE}| = 0.7\,\text{V}$ and once with each transistor's $|V_{BE}|$ changed to 0.6 and 0.8 V:

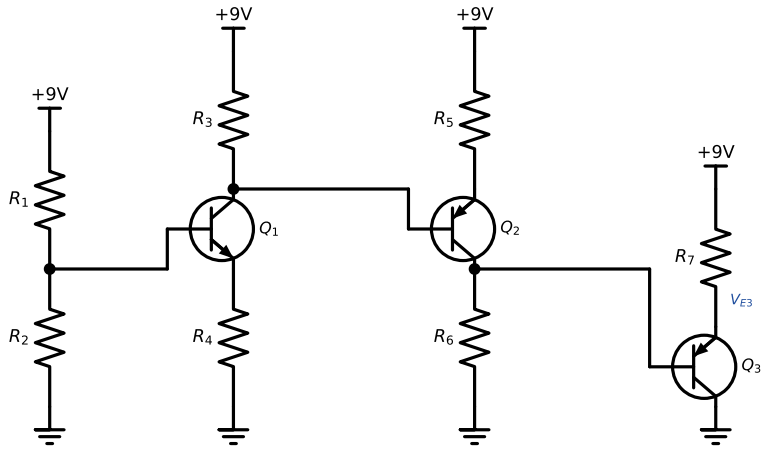
	target I_C	exact I_C	$\lvert V_{BE}\rvert = 0.6$	$\lvert V_{BE}\rvert = 0.8$	max change
Q_1	2 mA	1.94 mA	2.13 mA	1.74 mA	9.7 %
Q_2	4 mA	3.63 mA	4.01 mA	3.24 mA	9.6 %
Q_3	6 mA	6.27 mA	6.51 mA	6.04 mA	3.9 %

A ± 0.1 V change of V_{BE} moves no current by more than about 10 % of its design value (V_{BE}-insensitive), and the exact currents are close to the targets (I_B negligible). The exact currents drift further from the targets along a chain of direct-coupled stages: each base current and each $I_C = \alpha I_E$ slightly shifts the collector voltage that biases the next stage, and with only ≈ 1 V across the next emitter resistor that shift is amplified. This is the normal accuracy of a hand design with $I_B \approx 0$; a larger emitter-resistor voltage in the later stages reduces the drift.

Answer

$R_1 = 3.3\,\text{k}\Omega, \quad R_2 = 1.7\,\text{k}\Omega, \quad R_3 = 850\,\Omega, \quad R_4 = 500\,\Omega, \quad R_5 = 250\,\Omega, \quad R_6 = 450\,\Omega, \quad R_7 = 417\,\Omega$

Solution 4C.14 **Hard**



Step 1 — Rules

With $I_B \approx 0$ and every transistor active ($|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$), a resistor is known as soon as the voltages at both of its ends and the current through it are known: $R = \Delta V/I$. The specifications give the currents and some node voltages; the remaining freedom is removed with two rules:

- V_{BE}-insensitive: the voltage across the resistor that sets an emitter current must be at least 1 V (then $\pm 0.1\,\text{V}$ of V_{BE} changes I_C by at most 10 %);
- β-insensitive: a divider current must be much larger ($\geq 10\times$) than the base current it feeds, and a collector current that drives a base must be much larger than that base current.

Work from the nodes that are already fixed towards the unknown ones.

Step 2 — Design, node by node

1. V_{BE}-insensitivity: a 0.1 V change of V_{BE} changes the voltage across R_5 by 0.1 V, so $\frac{\Delta I_C}{I_C} = \frac{0.1}{V_{R5}} \leq 10\%$ needs $V_{R5} \geq 1\,\text{V}$. Choose the smallest value, $V_{R5} = 1\,\text{V}$: $V_{E2} = 9 - V_{R5} = 9 - 1 = 8\,\text{V}$.
2. V_{BE}-insensitivity: a 0.1 V change of V_{BE} changes the voltage across R_4 by 0.1 V, so $\frac{\Delta I_C}{I_C} = \frac{0.1}{V_{R4}} \leq 10\%$ needs $V_{R4} \geq 1\,\text{V}$. Choose the smallest value, $V_{R4} = 1\,\text{V}$: $V_{E1} = 0 + V_{R4} = 0 + 1 = 1\,\text{V}$.
3. $I_{dc} = I_{R1} + I_{C1} + I_{C2} + I_{C3} \leq 9\,\text{mA}$ leaves at most $9 - 1 - 2 - 5 = 1\,\text{mA}$ for the divider, and I_{R2} must be $\gg I_{B1} = 10\,\mu\text{A}$. Choose $I_{R2} = 0.5\,\text{mA}$.
4. Q_1 is active: $V_{B1} = V_{E1} + 0.7 = 1.7\,\text{V}$.
5. Q_2 is active: $V_{C1} = V_{E2} - 0.7 = 7.3\,\text{V}$.
6. Q_3 is active: $V_{C2} = V_{E3} - 0.7 = 4.3\,\text{V}$.
7. With $I_B \approx 0$ the only current through R_1 is I_{R2}: $I_{R1} = I_{R2} = 500\,\mu\text{A}$.
8. $R_1 = \dfrac{9 - V_{B1}}{I_{R2}} = \dfrac{9 - 1.7}{500\,\mu\text{A}} = \mathbf{14.6\,k\Omega}$
9. $R_2 = \dfrac{V_{B1} - 0}{I_{R2}} = \dfrac{1.7 - 0}{500\,\mu\text{A}} = \mathbf{3.4\,k\Omega}$
10. With $I_B \approx 0$ the only current through R_3 is I_{C1}: $I_{R3} = I_{C1} = 1\,\text{mA}$.

11. $R_3 = \dfrac{9 - V_{C1}}{I_{C1}} = \dfrac{9 - 7.3}{1\,\text{mA}} = \mathbf{1.7\,k\Omega}$

12. With $I_B \approx 0$ the only current through R_4 is I_{E1}: $I_{R4} = I_{E1} = 1\,\text{mA}$ ($I_C \approx I_E$).

13. $R_4 = \dfrac{V_{E1} - 0}{I_{E1}} = \dfrac{1 - 0}{1\,\text{mA}} = \mathbf{1\,k\Omega}$

14. With $I_B \approx 0$ the only current through R_5 is I_{E2}: $I_{R5} = I_{E2} = 2\,\text{mA}$ ($I_C \approx I_E$).

15. $R_5 = \dfrac{9 - V_{E2}}{I_{E2}} = \dfrac{9 - 8}{2\,\text{mA}} = \mathbf{500\,\Omega}$

16. With $I_B \approx 0$ the only current through R_6 is I_{C2}: $I_{R6} = I_{C2} = 2\,\text{mA}$.

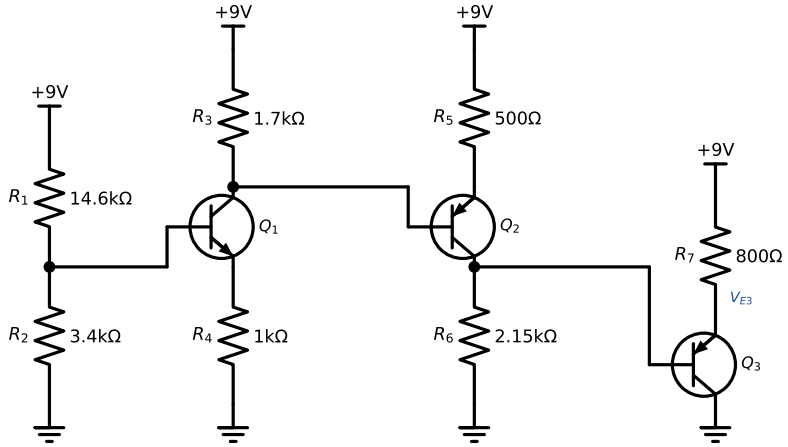
17. $R_6 = \dfrac{V_{C2} - 0}{I_{C2}} = \dfrac{4.3 - 0}{2\,\text{mA}} = \mathbf{2.15\,k\Omega}$

18. With $I_B \approx 0$ the only current through R_7 is I_{E3}: $I_{R7} = I_{E3} = 5\,\text{mA}$ ($I_C \approx I_E$).

19. $R_7 = \dfrac{9 - V_{E3}}{I_{E3}} = \dfrac{9 - 5}{5\,\text{mA}} = \mathbf{800\,\Omega}$

Step 3 — Designed circuit

resistor	value
R_1	14.6 kΩ
R_2	3.4 kΩ
R_3	1.7 kΩ
R_4	1 kΩ
R_5	500 Ω
R_6	2.15 kΩ
R_7	800 Ω



Step 4 — Check the design

- $I_{B1} = \frac{I_{C1}}{\beta} = 10\,\mu\text{A} \ll$ the 500 μA flowing in the base network ✓

- $I_{B2} = \frac{I_{C2}}{\beta} = 20\,\mu\text{A} \ll$ the 1 mA flowing in the base network ✓
- $I_{B3} = \frac{I_{C3}}{\beta} = 50\,\mu\text{A} \ll$ the 2 mA flowing in the base network ✓
- $V_{CE1} = V_{C1} - V_{E1} = 7.3 - 1 = 6.3\,\text{V} \geq 1\,\text{V}$ ✓
- $V_{EC2} = V_{E2} - V_{C2} = 8 - 4.3 = 3.7\,\text{V} \geq 1\,\text{V}$ ✓
- $V_{EC3} = V_{E3} - 0 = 5 - 0 = 5\,\text{V} \geq 1\,\text{V}$ ✓
- $I_{dc} = I_{R1} + I_{R3} + I_{R5} + I_{R7} = 500\,\mu\text{A} + 1\,\text{mA} + 2\,\text{mA} + 5\,\text{mA} = 8.5\,\text{mA} \leq 9\,\text{mA}$ ✓

Finally the designed circuit is solved exactly ($\beta = 100$, base currents included), once with $|V_{BE}| = 0.7\,\text{V}$ and once with each transistor's $|V_{BE}|$ changed to 0.6 and 0.8 V:

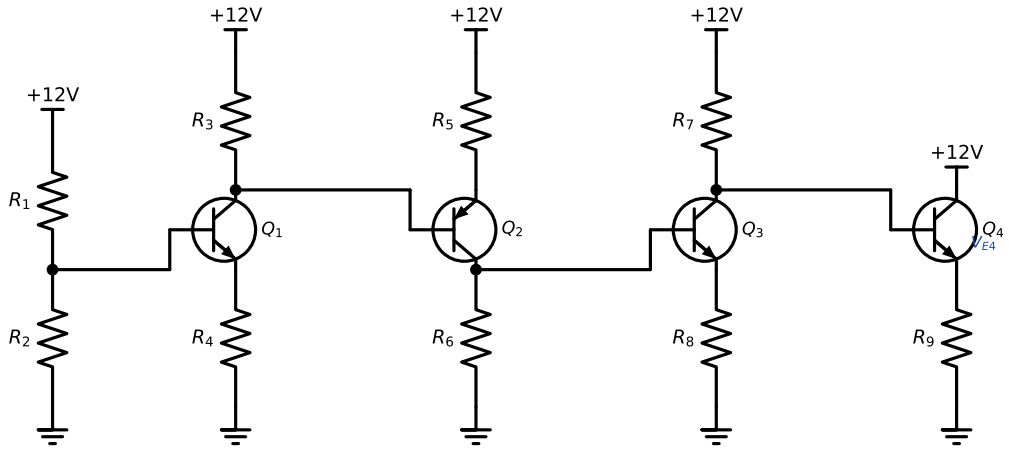
	target I_C	exact I_C	$\lvert V_{BE}\rvert = 0.6$	$\lvert V_{BE}\rvert = 0.8$	max change
Q_1	1 mA	964 µA	1.06 mA	867 µA	9.6 %
Q_2	2 mA	1.8 mA	1.99 mA	1.61 mA	9.6 %
Q_3	5 mA	5.35 mA	5.47 mA	5.23 mA	2.4 %

A ±0.1 V change of V_{BE} moves no current by more than about 10 % of its design value (V_{BE}-insensitive), and the exact currents are close to the targets (I_B negligible). The exact currents drift further from the targets along a chain of direct-coupled stages: each base current and each $I_C = \alpha I_E$ slightly shifts the collector voltage that biases the next stage, and with only ≈ 1 V across the next emitter resistor that shift is amplified. This is the normal accuracy of a hand design with $I_B \approx 0$; a larger emitter-resistor voltage in the later stages reduces the drift.

Answer

$R_1 = 14.6\,\text{k}\Omega$, $R_2 = 3.4\,\text{k}\Omega$, $R_3 = 1.7\,\text{k}\Omega$, $R_4 = 1\,\text{k}\Omega$, $R_5 = 500\,\Omega$, $R_6 = 2.15\,\text{k}\Omega$, $R_7 = 800\,\Omega$

Solution 4C.15 **Hard**



Step 1 — Rules

With $I_B \approx 0$ and every transistor active ($|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$), a resistor is known as soon as the voltages at both of its ends and the current through it are known: $R = \Delta V / I$. The specifications give the currents and some node voltages; the remaining freedom is removed with two rules:

- V_{BE}-insensitive: the voltage across the resistor that sets an emitter current must be at least 1 V (then $\pm 0.1\,\text{V}$ of V_{BE} changes I_C by at most 10 %);
- β-insensitive: a divider current must be much larger ($\geq 10\times$) than the base current it feeds, and a collector current that drives a base must be much larger than that base current.

Work from the nodes that are already fixed towards the unknown ones.

Step 2 — Design, node by node

1. V_{BE}-insensitivity: a 0.1 V change of V_{BE} changes the voltage across R_4 by 0.1 V, so $\frac{\Delta I_C}{I_C} = \frac{0.1}{V_{R4}} \leq 10\%$ needs $V_{R4} \geq 1\,\text{V}$. Choose the smallest value, $V_{R4} = 1\,\text{V}$: $V_{E1} = 0 + V_{R4} = 0 + 1 = 1\,\text{V}$.
2. V_{BE}-insensitivity: a 0.1 V change of V_{BE} changes the voltage across R_5 by 0.1 V, so $\frac{\Delta I_C}{I_C} = \frac{0.1}{V_{R5}} \leq 10\%$ needs $V_{R5} \geq 1\,\text{V}$. As specified (larger, for accuracy), $V_{R5} = 2\,\text{V}$: $V_{E2} = 12 - V_{R5} = 12 - 2 = 10\,\text{V}$.
3. V_{BE}-insensitivity: a 0.1 V change of V_{BE} changes the voltage across R_8 by 0.1 V, so $\frac{\Delta I_C}{I_C} = \frac{0.1}{V_{R8}} \leq 10\%$ needs $V_{R8} \geq 1\,\text{V}$. As specified (larger, for accuracy), $V_{R8} = 2\,\text{V}$: $V_{E3} = 0 + V_{R8} = 0 + 2 = 2\,\text{V}$.
4. The divider current must be much larger than the base current it feeds ($I_{B1} = 10\,\mu\text{A}$). Choose $I_{R2} = 500\,\mu\text{A}$ (50 times larger).
5. Q_1 is active: $V_{B1} = V_{E1} + 0.7 = 1.7\,\text{V}$.
6. Q_2 is active: $V_{C1} = V_{E2} - 0.7 = 9.3\,\text{V}$.
7. Q_3 is active: $V_{C2} = V_{E3} + 0.7 = 2.7\,\text{V}$.
8. Q_4 is active: $V_{C3} = V_{E4} + 0.7 = 6.7\,\text{V}$.
9. With $I_B \approx 0$ the only current through R_1 is I_{R2}: $I_{R1} = I_{R2} = 500\,\mu\text{A}$.
10. $R_1 = \dfrac{12 - V_{B1}}{I_{R2}} = \dfrac{12 - 1.7}{500\,\mu\text{A}} = \mathbf{20.6\,k\Omega}$

11. $R_2 = \dfrac{V_{B1} - 0}{I_{R2}} = \dfrac{1.7 - 0}{500\,\mu\text{A}} = \mathbf{3.4\,k\Omega}$

12. With $I_B \approx 0$ the only current through R_3 is I_{C1}: $I_{R3} = I_{C1} = 1\,\text{mA}$.

13. $R_3 = \dfrac{12 - V_{C1}}{I_{C1}} = \dfrac{12 - 9.3}{1\,\text{mA}} = \mathbf{2.7\,k\Omega}$

14. With $I_B \approx 0$ the only current through R_4 is I_{E1}: $I_{R4} = I_{E1} = 1\,\text{mA}$ ($I_C \approx I_E$).

15. $R_4 = \dfrac{V_{E1} - 0}{I_{E1}} = \dfrac{1 - 0}{1\,\text{mA}} = \mathbf{1\,k\Omega}$

16. With $I_B \approx 0$ the only current through R_5 is I_{E2}: $I_{R5} = I_{E2} = 2\,\text{mA}$ ($I_C \approx I_E$).

17. $R_5 = \dfrac{12 - V_{E2}}{I_{E2}} = \dfrac{12 - 10}{2\,\text{mA}} = \mathbf{1\,k\Omega}$

18. With $I_B \approx 0$ the only current through R_6 is I_{C2}: $I_{R6} = I_{C2} = 2\,\text{mA}$.

19. $R_6 = \dfrac{V_{C2} - 0}{I_{C2}} = \dfrac{2.7 - 0}{2\,\text{mA}} = \mathbf{1.35\,k\Omega}$

20. With $I_B \approx 0$ the only current through R_7 is I_{C3}: $I_{R7} = I_{C3} = 2\,\text{mA}$.

21. $R_7 = \dfrac{12 - V_{C3}}{I_{C3}} = \dfrac{12 - 6.7}{2\,\text{mA}} = \mathbf{2.65\,k\Omega}$

22. With $I_B \approx 0$ the only current through R_8 is I_{E3}: $I_{R8} = I_{E3} = 2\,\text{mA}$ ($I_C \approx I_E$).

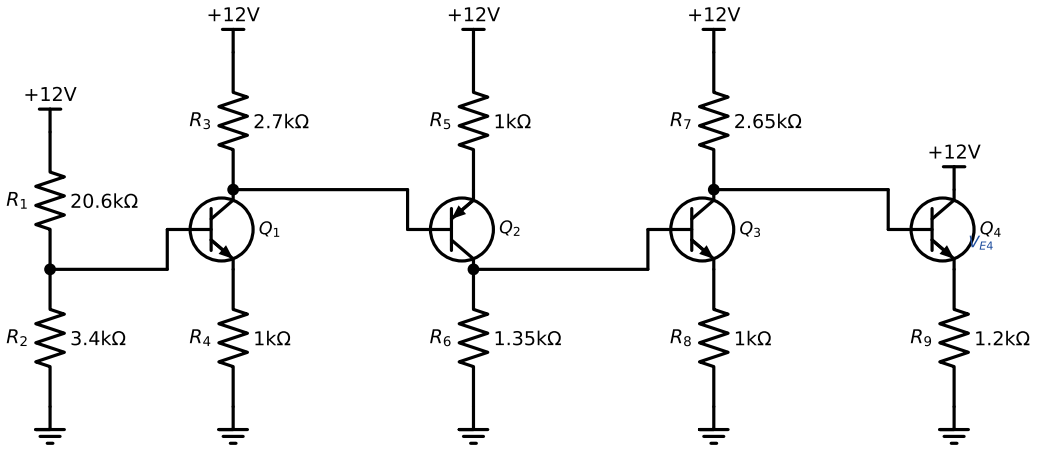
23. $R_8 = \dfrac{V_{E3} - 0}{I_{E3}} = \dfrac{2 - 0}{2\,\text{mA}} = \mathbf{1\,k\Omega}$

24. With $I_B \approx 0$ the only current through R_9 is I_{E4}: $I_{R9} = I_{E4} = 5\,\text{mA}$ ($I_C \approx I_E$).

25. $R_9 = \dfrac{V_{E4} - 0}{I_{E4}} = \dfrac{6 - 0}{5\,\text{mA}} = \mathbf{1.2\,k\Omega}$

Step 3 — Designed circuit

resistor	value
R_1	20.6 kΩ
R_2	3.4 kΩ
R_3	2.7 kΩ
R_4	1 kΩ
R_5	1 kΩ
R_6	1.35 kΩ
R_7	2.65 kΩ
R_8	1 kΩ
R_9	1.2 kΩ



Step 4 — Check the design

- $I_{B1} = \frac{I_{C1}}{\beta} = 10\,\mu\text{A} \ll$ the 500 μA flowing in the base network ✓
- $I_{B2} = \frac{I_{C2}}{\beta} = 20\,\mu\text{A} \ll$ the 1 mA flowing in the base network ✓
- $I_{B3} = \frac{I_{C3}}{\beta} = 20\,\mu\text{A} \ll$ the 2 mA flowing in the base network ✓
- $I_{B4} = \frac{I_{C4}}{\beta} = 50\,\mu\text{A} \ll$ the 2 mA flowing in the base network ✓
- $V_{CE1} = V_{C1} - V_{E1} = 9.3 - 1 = 8.3\,\text{V} \geq 1\,\text{V}$ ✓
- $V_{EC2} = V_{E2} - V_{C2} = 10 - 2.7 = 7.3\,\text{V} \geq 1\,\text{V}$ ✓
- $V_{CE3} = V_{C3} - V_{E3} = 6.7 - 2 = 4.7\,\text{V} \geq 1\,\text{V}$ ✓
- $V_{CE4} = 12 - V_{E4} = 12 - 6 = 6\,\text{V} \geq 1\,\text{V}$ ✓

Finally the designed circuit is solved exactly ($\beta = 100$, base currents included), once with $|V_{BE}| = 0.7\,\text{V}$ and once with each transistor's $|V_{BE}|$ changed to 0.6 and 0.8 V:

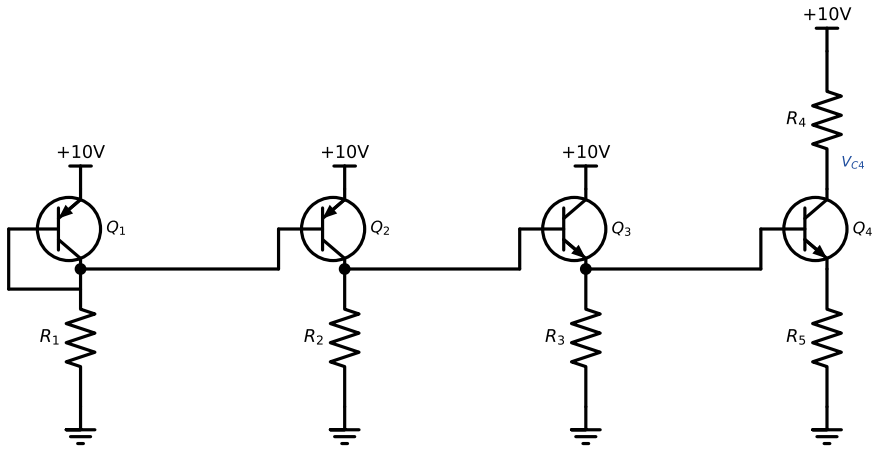
	target I_C	exact I_C	$\lvert V_{BE}\rvert = 0.6$	$\lvert V_{BE}\rvert = 0.8$	max change
Q_1	1 mA	962 μA	1.06 mA	866 μA	9.6 %
Q_2	2 mA	1.83 mA	1.93 mA	1.73 mA	4.8 %
Q_3	2 mA	1.73 mA	1.83 mA	1.63 mA	4.9 %
Q_4	5 mA	5.42 mA	5.5 mA	5.34 mA	1.6 %

A ± 0.1 V change of V_{BE} moves no current by more than about 10 % of its design value (V_{BE}-insensitive), and the exact currents are close to the targets (I_B negligible). The exact currents drift further from the targets along a chain of direct-coupled stages: each base current and each $I_C = \alpha I_E$ slightly shifts the collector voltage that biases the next stage, and with only ≈ 1 V across the next emitter resistor that shift is amplified. This is the normal accuracy of a hand design with $I_B \approx 0$; a larger emitter-resistor voltage in the later stages reduces the drift.

Answer

$R_1 = 20.6\,\mathrm{k\Omega}$, $R_2 = 3.4\,\mathrm{k\Omega}$, $R_3 = 2.7\,\mathrm{k\Omega}$, $R_4 = 1\,\mathrm{k\Omega}$, $R_5 = 1\,\mathrm{k\Omega}$, $R_6 = 1.35\,\mathrm{k\Omega}$, $R_7 = 2.65\,\mathrm{k\Omega}$, $R_8 = 1\,\mathrm{k\Omega}$, $R_9 = 1.2\,\mathrm{k\Omega}$

Solution 4C.16 **Hard**



Step 1 — Rules

With $I_B \approx 0$ and every transistor active ($|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$), a resistor is known as soon as the voltages at both of its ends and the current through it are known: $R = \Delta V / I$. The specifications give the currents and some node voltages; the remaining freedom is removed with two rules:

- V_{BE}-insensitive: the voltage across the resistor that sets an emitter current must be at least 1 V (then $\pm 0.1\,\text{V}$ of V_{BE} changes I_C by at most 10 %);
- β-insensitive: a divider current must be much larger ($\geq 10\times$) than the base current it feeds, and a collector current that drives a base must be much larger than that base current.

Work from the nodes that are already fixed towards the unknown ones.

Step 2 — Design, node by node

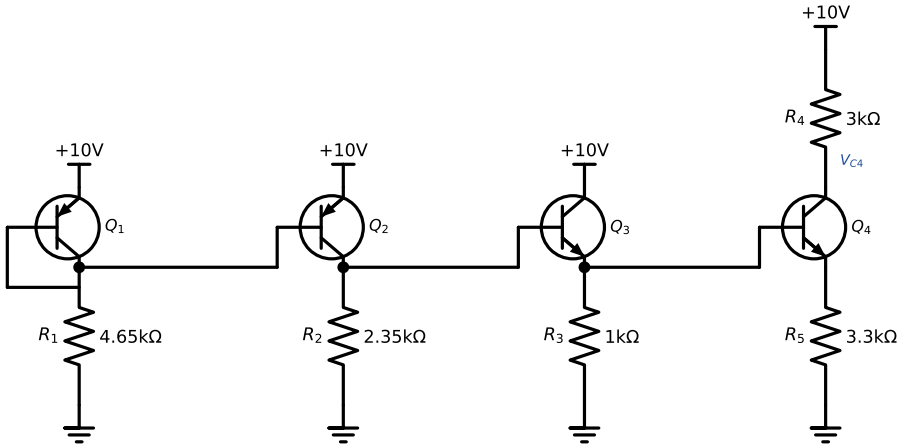
1. Q_1 (diode-connected) is active: $V_{B1} = 10 - 0.7 = 9.3\,\text{V}$.
2. Q_3 is active: $V_{C2} = V_{E3} + 0.7 = 4.7\,\text{V}$.
3. Q_4 is active: $V_{E4} = V_{E3} - 0.7 = 3.3\,\text{V}$.
4. Q_2 mirrors Q_1 (same base and emitter nodes): $I_{C2} = I_{C1} = 2\,\text{mA}$.
5. With $I_B \approx 0$ the only current through R_1 is I_{C1}: $I_{R1} = I_{C1} = 2\,\text{mA}$.
6. $R_1 = \dfrac{V_{B1} - 0}{I_{C1}} = \dfrac{9.3 - 0}{2\,\text{mA}} = \mathbf{4.65\,k\Omega}$
7. With $I_B \approx 0$ the only current through R_2 is I_{C2}: $I_{R2} = I_{C2} = 2\,\text{mA}$.
8. $R_2 = \dfrac{V_{C2} - 0}{I_{C2}} = \dfrac{4.7 - 0}{2\,\text{mA}} = \mathbf{2.35\,k\Omega}$
9. With $I_B \approx 0$ the only current through R_3 is I_{E3}: $I_{R3} = I_{E3} = 4\,\text{mA}$ ($I_C \approx I_E$).
10. $R_3 = \dfrac{V_{E3} - 0}{I_{E3}} = \dfrac{4 - 0}{4\,\text{mA}} = \mathbf{1\,k\Omega}$
11. With $I_B \approx 0$ the only current through R_4 is I_{C4}: $I_{R4} = I_{C4} = 1\,\text{mA}$.
12. $R_4 = \dfrac{10 - V_{C4}}{I_{C4}} = \dfrac{10 - 7}{1\,\text{mA}} = \mathbf{3\,k\Omega}$

13. With $I_B \approx 0$ the only current through R_5 is I_{E4}: $I_{R5} = I_{E4} = 1\,\text{mA}$ ($I_C \approx I_E$).

14. $R_5 = \dfrac{V_{E4} - 0}{I_{E4}} = \dfrac{3.3 - 0}{1\,\text{mA}} = \mathbf{3.3\,k\Omega}$

Step 3 — Designed circuit

resistor	value
R_1	$4.65\,\text{k}\Omega$
R_2	$2.35\,\text{k}\Omega$
R_3	$1\,\text{k}\Omega$
R_4	$3\,\text{k}\Omega$
R_5	$3.3\,\text{k}\Omega$



Step 4 — Check the design

- $I_{B1} = \frac{I_{C1}}{\beta} = 20\,\mu\text{A}$ (all bases on this node: $40\,\mu\text{A}$) $\ll$ the $2\,\text{mA}$ flowing in the base network ✓
- $I_{B2} = \frac{I_{C2}}{\beta} = 20\,\mu\text{A}$ (all bases on this node: $40\,\mu\text{A}$) $\ll$ the $2\,\text{mA}$ flowing in the base network ✓
- $I_{B3} = \frac{I_{C3}}{\beta} = 40\,\mu\text{A} \ll$ the $2\,\text{mA}$ flowing in the base network ✓
- $I_{B4} = \frac{I_{C4}}{\beta} = 10\,\mu\text{A} \ll$ the $4\,\text{mA}$ flowing in the base network ✓
- Q_1 is diode-connected: $V_{EC1} = |V_{BE}| = 0.7\,\text{V}$, always active (the $|V_{CE}|$ specification is meant for the other transistors) ✓
- $V_{EC2} = 10 - V_{C2} = 10 - 4.7 = 5.3\,\text{V} \geq 1\,\text{V}$ ✓
- $V_{CE3} = 10 - V_{E3} = 10 - 4 = 6\,\text{V} \geq 1\,\text{V}$ ✓
- $V_{CE4} = V_{C4} - V_{E4} = 7 - 3.3 = 3.7\,\text{V} \geq 1\,\text{V}$ ✓
- $I_{dc} = I_{R4} + I_{C1} + I_{C2} + I_{C3} = 1\,\text{mA} + 2\,\text{mA} + 2\,\text{mA} + 4\,\text{mA} = 9\,\text{mA} \leq 10\,\text{mA}$ ✓

Finally the designed circuit is solved exactly ($\beta = 100$, base currents included), once with $|V_{BE}| = 0.7\,\text{V}$ and once with each transistor's $|V_{BE}|$ changed to 0.6 and 0.8 V:

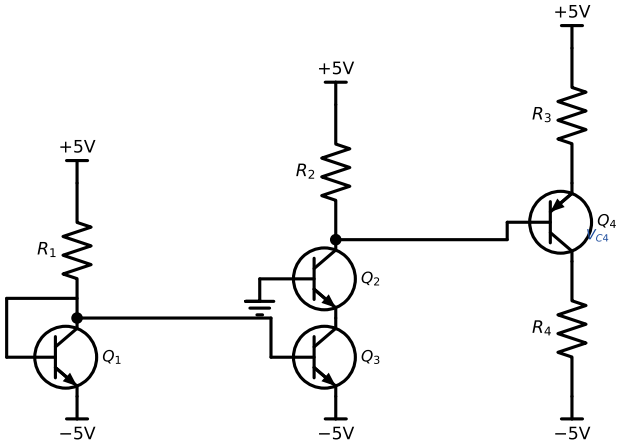
	target I_C	exact I_C	$\lvert V_{BE}\rvert = 0.6$	$\lvert V_{BE}\rvert = 0.8$	max change
Q_1	2 mA	1.96 mA	1.98 mA	1.94 mA	1.1 %
Q_2	2 mA	1.96 mA	1.98 mA	1.94 mA	1.1 %
Q_3	4 mA	3.79 mA	3.89 mA	3.69 mA	2.4 %
Q_4	1 mA	936 μA	966 μA	906 μA	3 %

A ± 0.1 V change of V_{BE} moves no current by more than about 10 % of its design value (V_{BE}-insensitive), and the exact currents are close to the targets (I_B negligible). The exact currents drift further from the targets along a chain of direct-coupled stages: each base current and each $I_C = \alpha I_E$ slightly shifts the collector voltage that biases the next stage, and with only ≈ 1 V across the next emitter resistor that shift is amplified. This is the normal accuracy of a hand design with $I_B \approx 0$; a larger emitter-resistor voltage in the later stages reduces the drift.

Answer

$R_1 = 4.65\,\text{k}\Omega, \quad R_2 = 2.35\,\text{k}\Omega, \quad R_3 = 1\,\text{k}\Omega, \quad R_4 = 3\,\text{k}\Omega, \quad R_5 = 3.3\,\text{k}\Omega$

Solution 4C.17 **Hard**



Step 1 — Rules

With $I_B \approx 0$ and every transistor active ($|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$), a resistor is known as soon as the voltages at both of its ends and the current through it are known: $R = \Delta V / I$. The specifications give the currents and some node voltages; the remaining freedom is removed with two rules:

- V_{BE}-insensitive: the voltage across the resistor that sets an emitter current must be at least 1 V (then $\pm 0.1\,\text{V}$ of V_{BE} changes I_C by at most 10 %);
- β-insensitive: a divider current must be much larger ($\geq 10\times$) than the base current it feeds, and a collector current that drives a base must be much larger than that base current.

Work from the nodes that are already fixed towards the unknown ones.

Step 2 — Design, node by node

1. V_{BE}-insensitivity: a 0.1 V change of V_{BE} changes the voltage across R_3 by 0.1 V, so $\frac{\Delta I_C}{I_C} = \frac{0.1}{V_{R3}} \leq 10\%$ needs $V_{R3} \geq 1\,\text{V}$. Choose the smallest value, $V_{R3} = 1\,\text{V}$: $V_{E4} = 5 - V_{R3} = 5 - 1 = 4\,\text{V}$.
2. Q_1 (diode-connected) is active: $V_{B1} = (-5) + 0.7 = -4.3\,\text{V}$.
3. Q_2 is active: $V_{E2} = 0 - 0.7 = -0.7\,\text{V}$.
4. Q_4 is active: $V_{C2} = V_{E4} - 0.7 = 3.3\,\text{V}$.
5. Q_3 mirrors Q_1 (same base and emitter nodes): $I_{C3} = I_{C1} = 1\,\text{mA}$.
6. KCL at V_{E2} ($I_B \approx 0$): $I_{E2} = I_{C3} = 1\,\text{mA} \approx I_{C2}$.
7. With $I_B \approx 0$ the only current through R_1 is I_{C1}: $I_{R1} = I_{C1} = 1\,\text{mA}$.
8. $R_1 = \dfrac{5 - V_{B1}}{I_{C1}} = \dfrac{5 - (-4.3)}{1\,\text{mA}} = \mathbf{9.3\,k\Omega}$
9. With $I_B \approx 0$ the only current through R_2 is I_{C2}: $I_{R2} = I_{C2} = 1\,\text{mA}$.
10. $R_2 = \dfrac{5 - V_{C2}}{I_{C2}} = \dfrac{5 - 3.3}{1\,\text{mA}} = \mathbf{1.7\,k\Omega}$
11. With $I_B \approx 0$ the only current through R_3 is I_{E4}: $I_{R3} = I_{E4} = 2\,\text{mA}$ ($I_C \approx I_E$).

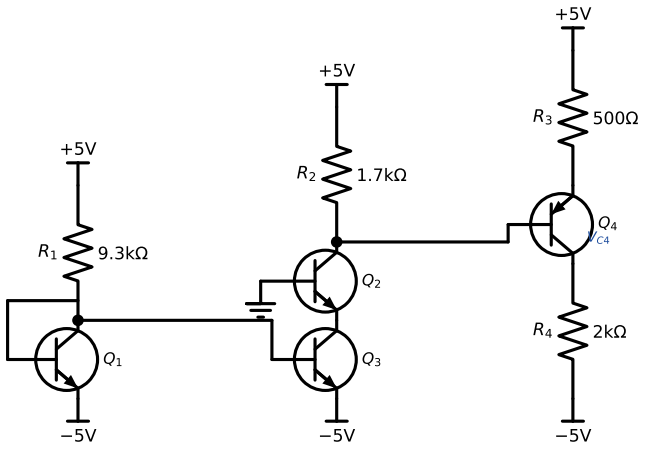
12. $R_3 = \dfrac{5 - V_{E4}}{I_{E4}} = \dfrac{5-4}{2\,\text{mA}} = \mathbf{500\,\Omega}$

13. With $I_B \approx 0$ the only current through R_4 is I_{C4}: $I_{R4} = I_{C4} = 2\,\text{mA}$.

14. $R_4 = \dfrac{V_{C4} - (-5)}{I_{C4}} = \dfrac{-1-(-5)}{2\,\text{mA}} = \mathbf{2\,k\Omega}$

Step 3 — Designed circuit

resistor	value
R_1	$9.3\,\text{k}\Omega$
R_2	$1.7\,\text{k}\Omega$
R_3	$500\,\Omega$
R_4	$2\,\text{k}\Omega$



Step 4 — Check the design

- $I_{B1} = \frac{I_{C1}}{\beta} = 10\,\mu\text{A}$ (all bases on this node: $20\,\mu\text{A}$) $\ll$ the $1\,\text{mA}$ flowing in the base network ✓
- $I_{B3} = \frac{I_{C3}}{\beta} = 10\,\mu\text{A}$ (all bases on this node: $20\,\mu\text{A}$) $\ll$ the $1\,\text{mA}$ flowing in the base network ✓
- $I_{B4} = \frac{I_{C4}}{\beta} = 20\,\mu\text{A} \ll$ the $1\,\text{mA}$ flowing in the base network ✓
- Q_1 is diode-connected: $V_{CE1} = |V_{BE}| = 0.7\,\text{V}$, always active (the $|V_{CE}|$ specification is meant for the other transistors) ✓
- $V_{CE2} = V_{C2} - V_{E2} = 3.3 - (-0.7) = 4\,\text{V} \geq 1\,\text{V}$ ✓
- $V_{CE3} = V_{E2} - (-5) = -0.7 - (-5) = 4.3\,\text{V} \geq 1\,\text{V}$ ✓
- $V_{EC4} = V_{E4} - V_{C4} = 4 - (-1) = 5\,\text{V} \geq 1\,\text{V}$ ✓

Finally the designed circuit is solved exactly ($\beta = 100$, base currents included), once with $|V_{BE}| = 0.7\,\text{V}$ and once with each transistor's $|V_{BE}|$ changed to 0.6 and $0.8\,\text{V}$:

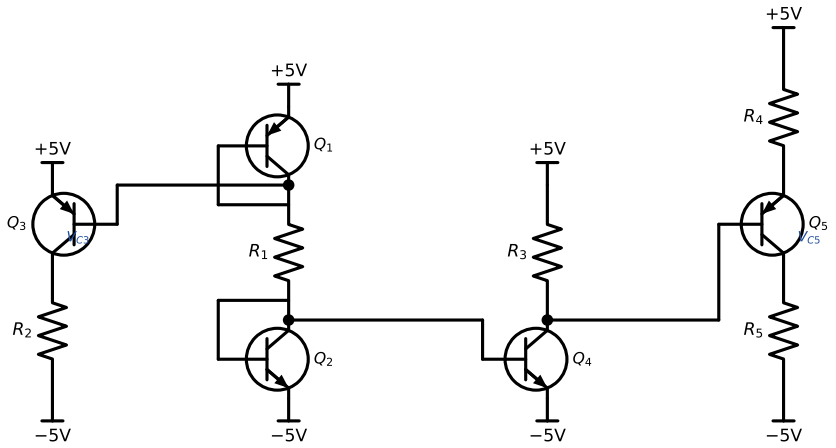
	target I_C	exact I_C	$\lvert V_{BE}\rvert = 0.6$	$\lvert V_{BE}\rvert = 0.8$	max change
Q_1	1 mA	980 µA	991 µA	970 µA	1.1 %
Q_2	1 mA	971 µA	971 µA	971 µA	0 %
Q_3	1 mA	980 µA	991 µA	970 µA	1.1 %
Q_4	2 mA	1.82 mA	2.01 mA	1.63 mA	9.6 %

A ± 0.1 V change of V_{BE} moves no current by more than about 10 % of its design value (V_{BE}-insensitive), and the exact currents are close to the targets (I_B negligible). The exact currents drift further from the targets along a chain of direct-coupled stages: each base current and each $I_C = \alpha I_E$ slightly shifts the collector voltage that biases the next stage, and with only ≈ 1 V across the next emitter resistor that shift is amplified. This is the normal accuracy of a hand design with $I_B \approx 0$; a larger emitter-resistor voltage in the later stages reduces the drift.

Answer

$R_1 = 9.3\,\text{k}\Omega, \quad R_2 = 1.7\,\text{k}\Omega, \quad R_3 = 500\,\Omega, \quad R_4 = 2\,\text{k}\Omega$

Solution 4C.18 **Hard**



Step 1 — Rules

With $I_B \approx 0$ and every transistor active ($|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$), a resistor is known as soon as the voltages at both of its ends and the current through it are known: $R = \Delta V / I$. The specifications give the currents and some node voltages; the remaining freedom is removed with two rules:

- V_{BE}-insensitive: the voltage across the resistor that sets an emitter current must be at least 1 V (then $\pm 0.1\,\text{V}$ of V_{BE} changes I_C by at most 10 %);
- β-insensitive: a divider current must be much larger ($\geq 10\times$) than the base current it feeds, and a collector current that drives a base must be much larger than that base current.

Work from the nodes that are already fixed towards the unknown ones.

Step 2 — Design, node by node

1. V_{BE}-insensitivity: a 0.1 V change of V_{BE} changes the voltage across R_4 by 0.1 V, so $\frac{\Delta I_C}{I_C} = \frac{0.1}{V_{R4}} \leq 10\%$ needs $V_{R4} \geq 1\,\text{V}$. Choose the smallest value, $V_{R4} = 1\,\text{V}$: $V_{E5} = 5 - V_{R4} = 5 - 1 = 4\,\text{V}$.
2. Q_3 is active: $V_{B3} = 5 - 0.7 = 4.3\,\text{V}$.
3. Q_2 (diode-connected) is active: $V_{B2} = (-5) + 0.7 = -4.3\,\text{V}$.
4. Q_5 is active: $V_{C4} = V_{E5} - 0.7 = 3.3\,\text{V}$.
5. Q_3 mirrors Q_1 (same base and emitter nodes): $I_{C3} = I_{C1} = 1\,\text{mA}$.
6. With $I_B \approx 0$ the only current through R_2 is I_{C3}: $I_{R2} = I_{C3} = 1\,\text{mA}$.
7. $R_2 = \dfrac{V_{C3} - (-5)}{I_{C3}} = \dfrac{2 - (-5)}{1\,\text{mA}} = \mathbf{7\,k\Omega}$
8. With $I_B \approx 0$ the only current through R_1 is I_{C1}: $I_{R1} = I_{C1} = 1\,\text{mA}$.
9. $R_1 = \dfrac{V_{B3} - V_{B2}}{I_{C1}} = \dfrac{4.3 - (-4.3)}{1\,\text{mA}} = \mathbf{8.6\,k\Omega}$
10. KCL at V_{B2} ($I_B \approx 0$): $I_{C2} = I_{C1} = 1\,\text{mA}$.
11. Q_4 mirrors Q_2 (same base and emitter nodes): $I_{C4} = I_{C2} = 1\,\text{mA}$.
12. With $I_B \approx 0$ the only current through R_3 is I_{C4}: $I_{R3} = I_{C4} = 1\,\text{mA}$.

13. $R_3 = \dfrac{5 - V_{C4}}{I_{C4}} = \dfrac{5 - 3.3}{1\,\text{mA}} = \mathbf{1.7\,k\Omega}$

14. With $I_B \approx 0$ the only current through R_4 is I_{E5}: $I_{R4} = I_{E5} = 2\,\text{mA}$ ($I_C \approx I_E$).

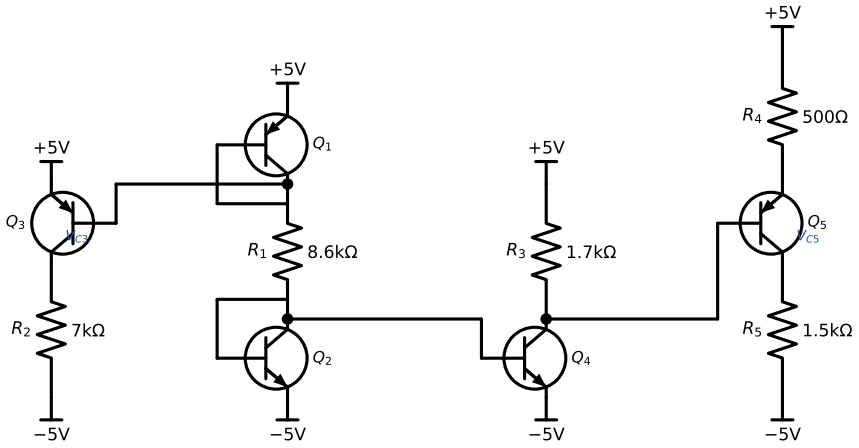
15. $R_4 = \dfrac{5 - V_{E5}}{I_{E5}} = \dfrac{5 - 4}{2\,\text{mA}} = \mathbf{500\,\Omega}$

16. With $I_B \approx 0$ the only current through R_5 is I_{C5}: $I_{R5} = I_{C5} = 2\,\text{mA}$.

17. $R_5 = \dfrac{V_{C5} - (-5)}{I_{C5}} = \dfrac{-2 - (-5)}{2\,\text{mA}} = \mathbf{1.5\,k\Omega}$

Step 3 — Designed circuit

resistor	value
R_1	8.6 kΩ
R_2	7 kΩ
R_3	1.7 kΩ
R_4	500 Ω
R_5	1.5 kΩ



Step 4 — Check the design

- $I_{B3} = \frac{I_{C3}}{\beta} = 10\,\mu\text{A}$ (all bases on this node: $20\,\mu\text{A}$) $\ll$ the 1 mA flowing in the base network ✓
- $I_{B1} = \frac{I_{C1}}{\beta} = 10\,\mu\text{A}$ (all bases on this node: $20\,\mu\text{A}$) $\ll$ the 1 mA flowing in the base network ✓
- $I_{B2} = \frac{I_{C2}}{\beta} = 10\,\mu\text{A}$ (all bases on this node: $20\,\mu\text{A}$) $\ll$ the 1 mA flowing in the base network ✓
- $I_{B4} = \frac{I_{C4}}{\beta} = 10\,\mu\text{A}$ (all bases on this node: $20\,\mu\text{A}$) $\ll$ the 1 mA flowing in the base network ✓
- $I_{B5} = \frac{I_{C5}}{\beta} = 20\,\mu\text{A} \ll$ the 1 mA flowing in the base network ✓
- $V_{EC3} = 5 - V_{C3} = 5 - 2 = 3\,\text{V} \geq 1\,\text{V}$ ✓
- Q_1 is diode-connected: $V_{EC1} = |V_{BE}| = 0.7\,\text{V}$, always active (the $|V_{CE}|$ specification is meant for the other transistors) ✓

- Q_2 is diode-connected: $V_{CE2} = |V_{BE}| = 0.7\,\text{V}$, always active (the $|V_{CE}|$ specification is meant for the other transistors) ✓
- $V_{CE4} = V_{C4} - (-5) = 3.3 - (-5) = 8.3\,\text{V} \geq 1\,\text{V}$ ✓
- $V_{EC5} = V_{E5} - V_{C5} = 4 - (-2) = 6\,\text{V} \geq 1\,\text{V}$ ✓

Finally the designed circuit is solved exactly ($\beta = 100$, base currents included), once with $|V_{BE}| = 0.7\,\text{V}$ and once with each transistor's $|V_{BE}|$ changed to 0.6 and 0.8 V:

	target I_C	exact I_C	$\lvert V_{BE}\rvert = 0.6$	$\lvert V_{BE}\rvert = 0.8$	max change
Q_3	1 mA	980 µA	992 µA	969 µA	1.1 %
Q_1	1 mA	980 µA	992 µA	969 µA	1.1 %
Q_2	1 mA	980 µA	992 µA	969 µA	1.1 %
Q_4	1 mA	980 µA	992 µA	969 µA	1.1 %
Q_5	2 mA	1.85 mA	2.04 mA	1.66 mA	9.6 %

A $\pm 0.1\,\text{V}$ change of V_{BE} moves no current by more than about 10 % of its design value (V_{BE}-insensitive), and the exact currents are close to the targets (I_B negligible). The exact currents drift further from the targets along a chain of direct-coupled stages: each base current and each $I_C = \alpha I_E$ slightly shifts the collector voltage that biases the next stage, and with only $\approx 1\,\text{V}$ across the next emitter resistor that shift is amplified. This is the normal accuracy of a hand design with $I_B \approx 0$; a larger emitter-resistor voltage in the later stages reduces the drift.

Answer

$R_1 = 8.6\,\text{k}\Omega, \quad R_2 = 7\,\text{k}\Omega, \quad R_3 = 1.7\,\text{k}\Omega, \quad R_4 = 500\,\Omega, \quad R_5 = 1.5\,\text{k}\Omega$